

EEG

FINAL REPORT

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1 Abstract

Electroencephalography (EEG) provides an exceptional temporal resolution for monitoring continuous cortical voltage fluctuations, yet its signal fidelity is frequently degraded by physiological artifacts, environmental noise, and technical limitations such as low-impedance electrolyte salt bridges. This investigation addresses these continuous monitoring challenges through a dual-phase quantitative framework encompassing automated noise and artifact detection alongside predictive machine learning. In the first phase, the four primary physiological neuro-oscillated bands- Delta (0.5 – 4 Hz), Theta (4 – 8 Hz), Alpha (8 – 13 Hz), and Beta (13 – 30 Hz)-were isolated across shifting ocular states, demonstrating a prominent increase in occipital alpha-band standard deviation during eye closure ($12.21 \mu V$) compared to relaxed wakefulness ($3.68 \mu V$). To screen for recording compromises, an automated Electrical Distance (*ED*) algorithm detected localized electrolyte bridging artifacts, identifying the *P8 – T8* channel configuration as the primary short-circuit risk with a minimal normalized *ED* of 0.1771. In the second phase, Independent Component Analysis (ICA) coupled with the ICLabel extension statistically separated cortical sources from non-neural ocular (99.7% probability) and cardiac (96.4% probability) artifact pathways. Spectral features were subsequently extracted using a modified Welch periodogram across two central-parietal independent neural generators to train a linear Support Vector Machine (SVM) for continuous imagined emotion decoding (Happy vs. Sad). Following Leave-One-Out Cross-Validation (LOOCV) and operating threshold calibration, the classifier successfully achieved a high specificity of 79.17% and an operational Area Under the Curve (AUC) of 0.69. These findings validate the technical integration of statistical artifact rejection, electrical distance topography screening, and optimized machine learning frameworks for advanced biomedical engineering applications in recognizing internal neural states.

2 Introduction

Electroencephalography (EEG) has emerged as a pivotal non-invasive neuroimaging modality utilized extensively to capture the complex electrical activity of the human brain with exceptional temporal resolution [1]. The continuous monitoring of postsynaptic potentials from the scalp allows for the

direct observation of distinct neural oscillations, which are traditionally categorized into specific frequency bands including alpha (α), beta (β), delta (δ), and theta (θ) waves. These oscillatory patterns serve as critical indicators of diverse physiological and cognitive states [1]. For instance, delta waves (0.5 – 4 Hz) are predominately associated with deep sleep phases, while theta waves (4 – 8 Hz) are observed during periods of drowsiness and memory processing. Concurrently, alpha oscillations (8 – 12 Hz) are strongly correlated with relaxed wakefulness, famously modulating in amplitude upon the closing and opening of the eyes, whereas beta waves (12 – 30 Hz) signify active cognitive engagement, focus, and motor preparation.

Despite the profound diagnostic and investigative utility of EEG, the acquired biological signals are highly susceptible to multiple classes of physiological and environmental noise [1]. In practical laboratory settings, standard signal acquisition is frequently compromised by ocular artifacts such as blinks and saccades, alongside technical distortions including muscle activity and electrical interference. Furthermore, dense electrode montages are uniquely prone to the "salt bridge" artifact, a phenomenon wherein electrolyte over-application or perspiration creates a low-impedance electrical short between neighboring channels [2]. This short-circuiting effect introduces an artificial similarity between the waveforms of adjacent electrodes, significantly distorting the spatial topography of the underlying cortical potentials and undermining the validity of subsequent classifications [2]. To preserve the integrity of neural data, mathematical source separation techniques, most notably Independent Component Analysis (ICA), are widely deployed to isolate true physiological generators from non-neural noise artifacts [1, 3].

While extensive literature has characterized the properties of executing motor commands or processing static cognitive tasks, fewer investigations have integrated both automated electrical distance assessments for artifact detection and advanced machine learning frameworks for the continuous decoding of internal emotional states using premovement independent neural sources [2, 3]. This experimental investigation sought to address these interconnected gaps by implementing a dual-phase analysis of multi-channel EEG data. Therefore, the purpose of this study was to isolate and evaluate the distinct properties of the four primary neuro-oscillatory waves under changing ocular states, systematically detect the occurrence and timing of localized electrolyte bridges using electrical distance distributions [2], and develop a robust predictive classifier capable of decoding distinct neural intentions from isolated premovement independent components [3].

3 Materials and methods

Part 1: Signal Acquisition, Preprocessing, and Electrolyte Bridge Detection

Data Acquisition and Experimental Protocol

The experimental procedure utilized the Emotiv EPOC+ high-density electroencephalography (EEG) system to capture biological voltage fluctuations across the scalp. The data recording environment active throughout the investigation consisted of a wireless 14-channel electrode montage structurally positioned in conformity with the international 10-20 system. Prior to placement, the felt pads of the electrodes were hydrated using tap water to establish sufficient electrical conductivity between the sensors and the scalp. Cortical electrical potentials were captured at a fixed hardware sampling frequency denoted as f_s . The discrete sampling time interval, denoted as T_s , was mathematically defined using the following relation:

$$(1) T_s [\text{sec}] = \frac{1}{f_s [\text{Hz}]}$$

Utilizing this sampling interval, the maximum sample length, which represented the total cumulative duration of the recorded experimental block (T_{max}), was determined as follows:

$$(2) T_{max} [\text{sec}] = (\text{number of samples} - 1) \cdot T_s [\text{sec}]$$

The experimental timeline was structured into three sequential observation phases to capture baseline dynamics under shifting visual inputs: an initial baseline phase with eyes open, a subsequent phase with eyes closed, and a concluding phase with eyes open.

Preprocessing and Statistical Identification of Electrolyte Salt Bridges

To attenuate non-neural high-frequency environmental noise and eliminate low-frequency baseline drifts, the continuous 14-channel datasets were subjected to digital filtering. An offline 5th-order Butterworth band-pass filter was implemented with cut-off thresholds established between 0.5 Hz and 45 Hz. Power Spectral Density (PSD) analyses were executed across the isolated segments to measure the energy within the standard frequency spectra: delta (0.5 – 4 Hz), theta (4 – 8 Hz), alpha (8 – 12 Hz), and beta (12 – 30 Hz). The amplitude dispersion of these continuous waveforms was mathematically quantified using the Standard Deviation (STD).

To address and identify structural recording artifacts caused by the potential spreading or over-application of the hydration liquid between adjacent high-density scalp configurations, an automated algorithmic scanning process was executed. For any pair of neighboring recording channels, designated

as electrode i and electrode j , the difference signal $P_{ij}(t)$ between their electrical potentials over time was defined as:

$$(3) P_{ij}(t) [\mu V] = P_i(t) [\mu V] - P_j(t) [\mu V]$$

To quantify channel similarity, the Electrical Distance (ED) calculation was employed based on the average variance of this difference signal across T total samples:

$$(4) ED [\mu V^2] = \frac{1}{T} \sum_{t=1}^T \left(P_{ij}(t) - \overline{P_{ij}(t)} \right)^2$$

Where $\overline{P_{ij}(t)}$ represents the statistical mean value of the difference signal calculated over the evaluated epoch. An artificial drop in the calculated ED metric toward $0 \mu V^2$ verified the existence of a low-impedance electrolyte salt bridge.

Part 2: Source Separation and Emotion Predictive Classification

Independent Component Analysis and Artifact Extraction

Following initial spectral preprocessing, the filtered 14-channel EEG matrix was imported into the **EEGLAB software platform** to perform advanced source separation. Independent Component Analysis (ICA) was executed within the EEGLAB environment to separate the mixed scalp signals into statistically independent physiological components. Individual spatial heatmaps, temporal waveforms, and spectral profiles generated by the software were evaluated to classify the distinct independent components (ICs). Non-cortical artifact sources, specifically tracking ocular blink dynamics, vertical eye movements, and localized cardiac signals, were statistically isolated and systematically deleted from the clean signal matrix. Following artifact rejection, the purified cortical independent components corresponding to specific emotional stimulus epochs were extracted for feature derivation.

Feature Engineering and Predictive Classification Modeling

Machine learning matrices were assembled using the temporal properties of the isolated independent components. The analytical features were computed based on the Standard Deviation (STD) of the signal amplitude extracted across the filtered delta, theta, alpha, and beta frequency windows for the selected components. A supervised predictive machine learning classifier was trained and validated using an n -fold cross-validation scheme to automatically classify discrete internal emotional and cognitive states based on the computed independent component feature vectors.

To compile the feature matrix for emotion classification, a localized segmentation and spectral estimation pipeline was implemented. For each subject, a 1.0-second time window was advanced immediately following the logged time stamps of the emotional triggers (Happy and Sad button presses). Within each extracted epoch, power spectral density (PSD) estimation was conducted independently for each selected neural independent component using Welch's modified periodogram method (pwelch function in MATLAB). The Welch estimator was configured utilizing 0.5-second Hamming windows with a 50% overlap (0.25-second shift) and a 256-point Fast Fourier Transform (FFT) length, assuming a post-resampling frequency (f_s) of 100 [Hz]. Following the spectral transformation, the continuous frequency axis (f_axis) was bounded to extract the mean localized power ($\mu V^2/Hz$) across four distinct physiological neuro-oscillatory bands: Delta (0.5 – 4 Hz), Theta (4 – 8 Hz), Alpha (8 – 13 Hz), and Beta (13 – 30 Hz). This windowing and filtering architecture yielded a structured 8-dimensional feature vector per instance (4 frequency bands $\times$ 2 ICs), formalizing the quantitative input for the subsequent cross-validated linear SVM model.

To evaluate the mathematical performance of the cross-validated model, a Receiver Operating Characteristic (ROC) analysis was plotted across changing decision thresholds, and the Area Under the Curve (AUC) was computed. The discrete classification capabilities of the model were validated based on the mathematical metrics of Sensitivity and Specificity, which were derived from the baseline counts of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN):

$$(5) \text{ Sensitivity} = \frac{TP}{TP + FN}$$

$$(6) \text{ Specificity} = \frac{TN}{TN + FP}$$

The integrated AUC performance metric provided a threshold-independent measure of classifier efficacy in successfully decoding internal states from pre-processed neural projections.

4 Results

4.1 Part 1

In this part, the EEG signals were preprocessed and analyzed in both the time and frequency domains. The preprocessing stage included band-pass filtering, selection of representative electrodes, isolation of the four EEG rhythms, and calculation of the standard deviation and frequency characteristics of each wave.. Following the exclusion of noisy channels, seven electrodes remained for further analysis: FC5, O1, O2, P8, FC6, F4, and F8.

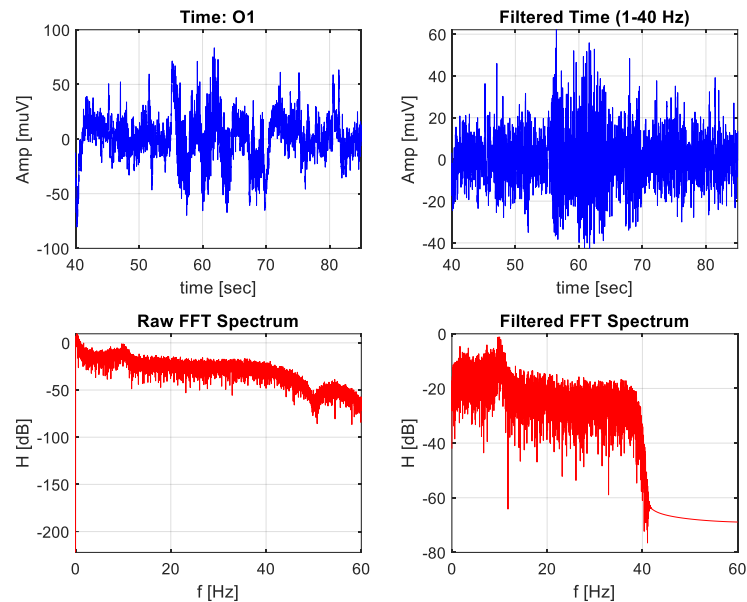


Figure 1: The O1 electrode before and after the application of the 1-40 Hz FIR bandpass filter.

The effectiveness of the preprocessing stage is demonstrated in Figure 1, which presents the O1 electrode before and after the application of the 1-40 Hz FIR bandpass filter. The upper panels show the signal in the time domain, while the lower panels present the corresponding frequency spectra. Before filtering, the signal contained substantial low-frequency drift and high-frequency noise. After filtering, the frequency spectrum was limited to the selected frequency range. Attenuation of frequencies below 1 Hz and above 40 Hz was observed. Figure 1 presents a representative example of the filtering procedure applied to the EEG signals. Similar improvements in signal quality were observed for all remaining electrodes following the preprocessing stage and are therefore not presented individually.

To identify the most representative electrode for each EEG rhythm, the spectral content of all remaining electrodes was examined. The electrode exhibiting the highest spectral activity within each EEG frequency band was selected for further analysis. The selected electrodes are presented in Table 1:

Wave	Selected Electrode
Delta (δ)	FC5
Theta (θ)	F8
Alpha (α)	O1
Beta (β)	F4

Table 1: Electrodes selected for the analysis of each EEG frequency band

The isolated waveforms obtained after applying the corresponding narrow band-pass filters are presented in Figures 2-5:

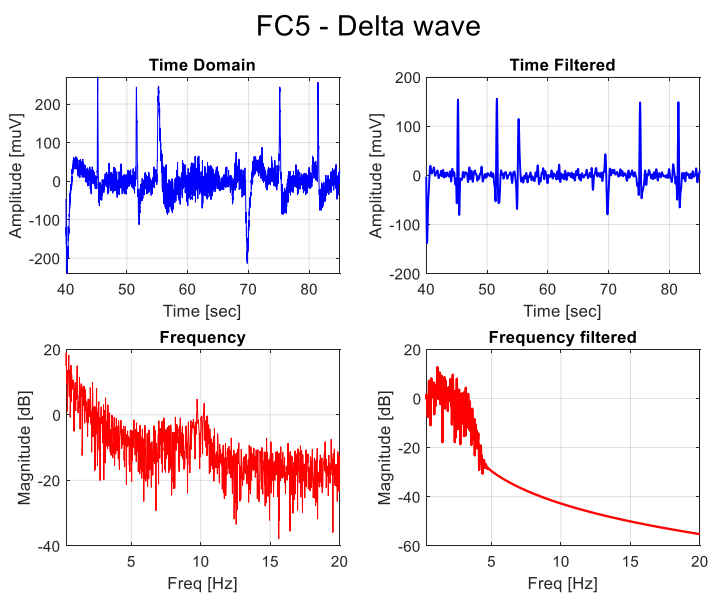


Figure 2: FC5 Delta wave

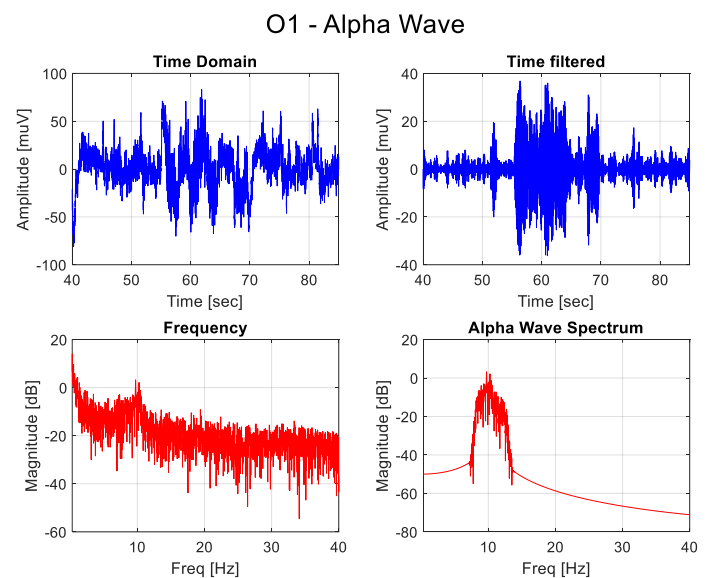


Figure 3: O1 Alpha wave

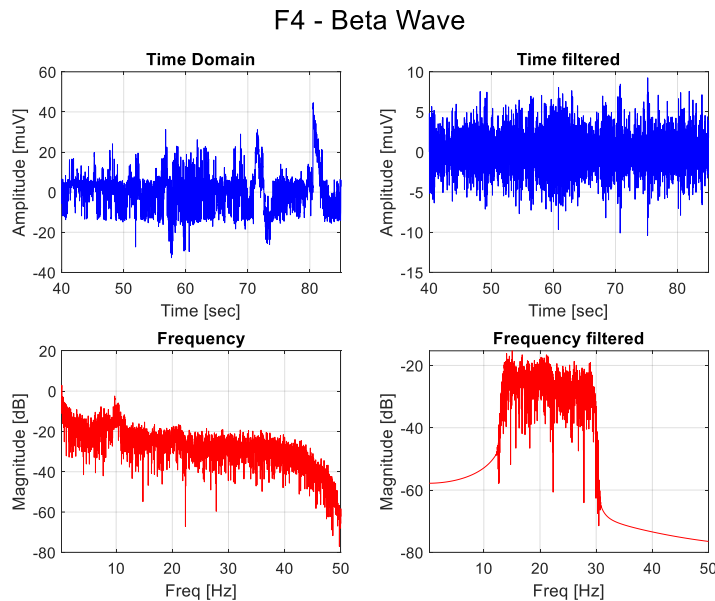


Figure 4: F4 Beta wave

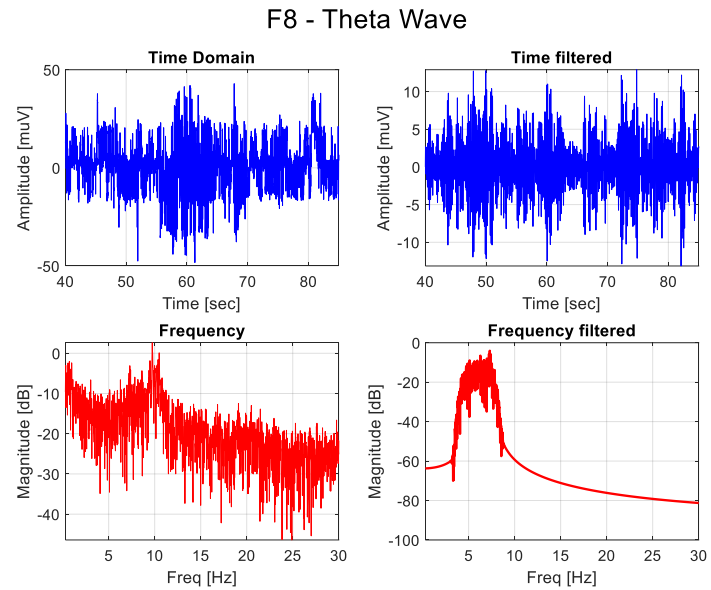


Figure 5: F8 Theta wave

Figures 2-5 present the isolated EEG rhythms obtained after applying the corresponding band-pass filters. Each figure presents the filtered EEG signal in the time domain together with its corresponding frequency spectrum.

The standard deviation values calculated for each wave and experimental segment are summarized in Table 2:

Wave	Eyes Open 1	Eyes Closed	Eyes Open 2
Alpha (α)	3.68	12.21	3.80
Beta (β)	4.78	5.77	5.33
Delta (δ)	7.50	5.92	6.17
Theta (θ)	6.88	6.07	6.09

Table 2: The standard deviation values calculated for each wave and experimental segment

The Alpha rhythm exhibited standard deviation values of 3.68 μV during Eyes Open 1, 12.21 μV during Eyes Closed, and 3.80 μV during Eyes Open 2. The Beta rhythm exhibited values ranging from 4.78 μV to 5.77 μV , whereas the Delta and Theta rhythms exhibited values between 5.92-7.50 μV and 6.07-6.88 μV , respectively.

The active frequency ranges calculated for each waveband are presented in Table 3:

Wave	Eyes Open 1 [Hz]	Eyes Closed [Hz]	Eyes Open 2 [Hz]	Average Range[Hz]
Alpha (α)	8.87–10.27	9.53–9.53	9.73–10.00	9.38–9.93
Beta (β)	13.67–20.73	14.53–20.27	13.93–16.60	14.04–19.20
Delta (δ)	1.33–2.00	1.07–2.20	0.67–2.13	1.02–2.11
Theta (θ)	7.13–7.20	5.07–7.20	5.40–7.40	5.87–7.27

Table 3: The active frequency ranges calculated for each waveband

The calculated active frequency ranges are presented in Table 3. The frequency ranges calculated for the four EEG rhythms remained within their corresponding frequency bands.

The mean frequencies calculated from the active frequency ranges are summarized in Table 4.

Wave	Eyes Open 1 [Hz]	Eyes Closed [Hz]	Eyes Open 2 [Hz]	Average [Hz]
Alpha (α)	9.62	9.53	9.85	9.67
Beta (β)	15.70	18.24	14.92	16.29
Delta (δ)	1.58	1.53	1.33	1.48
Theta (θ)	7.17	5.93	6.65	6.58

Table 4: The mean frequencies calculated from the active frequency ranges

The mean frequency values ranged from 1.48 Hz for the Delta rhythm to 16.29 Hz for the Beta rhythm. Small variations between the three recording segments were observed for the Alpha, Delta, and Theta rhythms, whereas larger differences were observed for the Beta rhythm.

4.2 Part 2 - Salt Bridge Detection

In this part, adjacent EEG electrode pairs were analyzed for the detection of possible salt bridge artifacts. The EEG signals recorded from each neighboring electrode pair were compared, and the absolute difference between the signals was calculated. In addition, the normalized Electrical Distance (ED) was calculated for each adjacent electrode pair.

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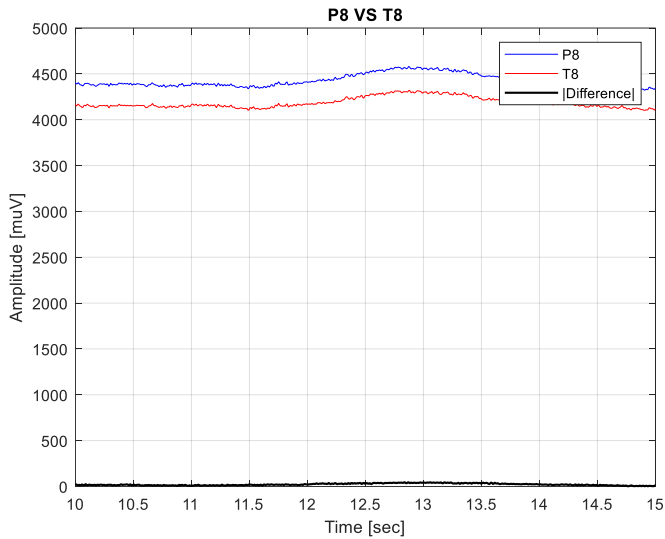


figure 6: P8 vs T8

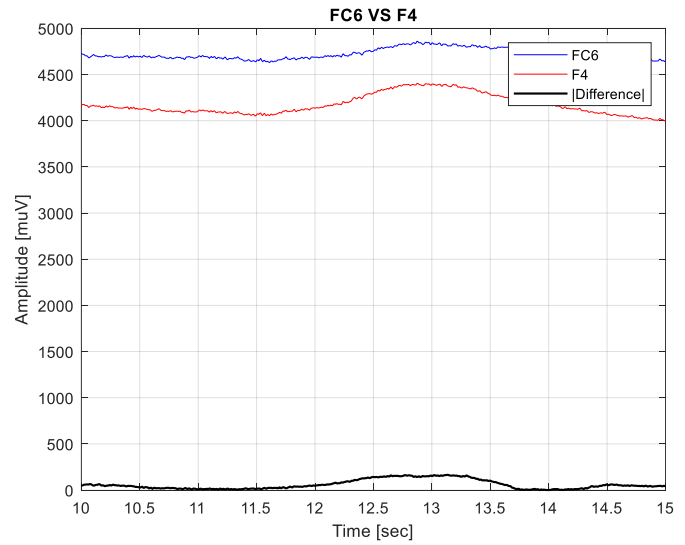


figure 7: FC6 vs F4

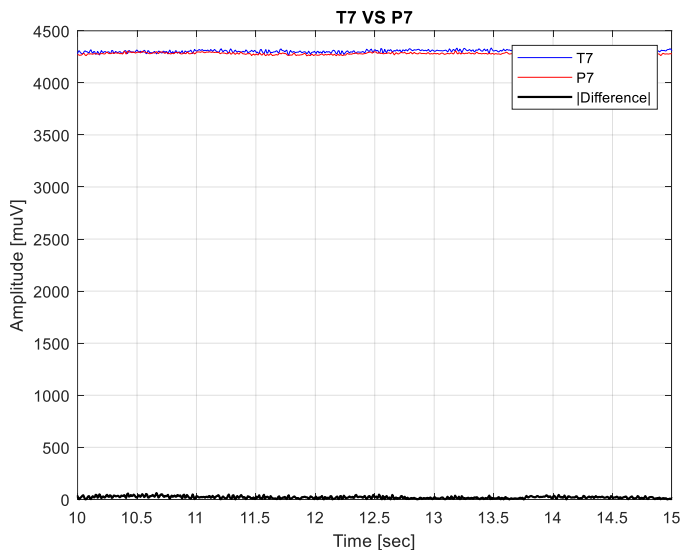


figure 8: T7 vs P7

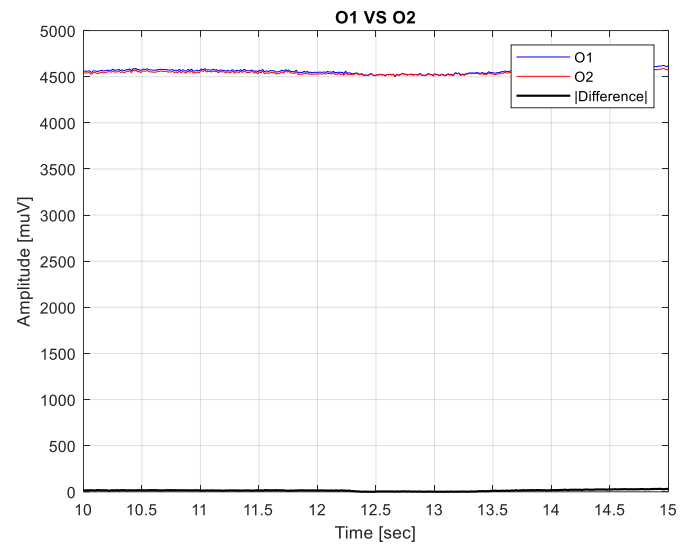


figure 9: O1 vs O2

Representative examples of the analyzed electrode pairs are presented in Figures 6-9. Figure 6 presents the EEG signals recorded from the P8 and T8 electrodes together with the absolute difference between the two signals. Figure 7 presents the EEG signals recorded from the FC6 and F4 electrodes together with the absolute difference between the two signals. Figure 8 presents the EEG signals recorded from the T7 and P7 electrodes together with the absolute difference between the two signals. Figure 9 presents the EEG signals recorded from the O1 and O2 electrodes together with the absolute difference between the two signals.

The normalized Electrical Distance values calculated for all adjacent electrode pairs are presented in Table 5:

Adjacent electrode pair	Normalized ED
AF3 - F7	1.8987
F7 - F3	3.7396
F3 - FC5	1.9293
FC5 - T7	2.2518
T7 - P7	0.8103
P7 - O1	1.8423
O1 - O2	1.2021
O2 - P8	1.5261
P8 - T8	0.1771
T8 - FC6	0.9314
FC6 - F4	0.5358
F4 - F8	1.1148
F8 - AF4	0.9982

Table 5: Normalized Electrical Distance (ED) values calculated for adjacent electrode pairs.

The normalized ED values ranged from 0.1771 to 3.7396. The lowest normalized ED value was obtained for the P8-T8 electrode pair (0.1771), followed by the FC6-F4 electrode pair (0.5358) and the T7-P7 electrode pair (0.8103). The highest normalized ED value was obtained for the F7-F3 electrode pair (3.7396).

4.3 Part 3 - Emotion Classification Results (Happy vs. Sad)

To isolate distinct neural activity from physiological noise, Independent Component Analysis (ICA) was performed across the multi-channel EEG recordings. Automated component classification using the ICLabel extension successfully identified non-neural artifact contributions.

The specific artifact signatures used for signal purification included:

Ocular Artifacts (Eye Blinks/Movements): Component 17 (and equivalent components across subjects) demonstrated a strong, highly localized anterior-frontal activation over the orbits. ICLabel assigned this

component an Eye artifact probability of 99.7%. (Representative isolated scalp topographic map is shown in Figure 11).

Cardiac Artifacts (ECG Contamination): Component 13 exhibited a characteristic asymmetrical, diagonal posterior peripheral topography typical of volume-conducted cardiac electrical activity across the scalp. This component was flagged with a Heart probability of 96.4%. (Representative isolated scalp topographic map is shown in Figure 11).

Following artifact profiling, two high-probability neural Independent Components (IC2 and IC3) displaying stable, bilateral central-parietal projections were selected for emotion feature extraction across all 5 subjects ($N = 5$). All other artifact-dominant components were mathematically removed from the datasets to isolate clean cortical tracking. To satisfy the layout guidelines, the full, uncropped 35-component EEGLAB matrices for all subjects were relegated to the Appendix to prevent text overcrowding.

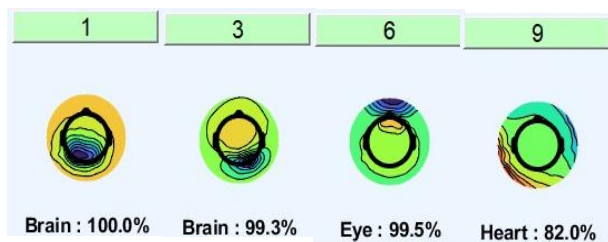


Figure 10: ICs for subject 4

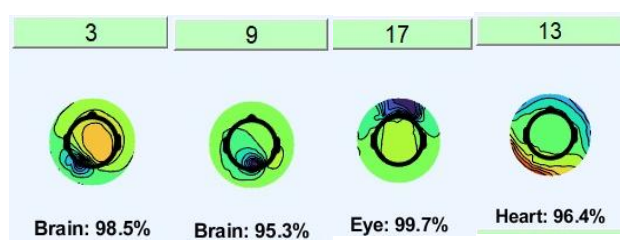


Figure 11: ICs for subject 10

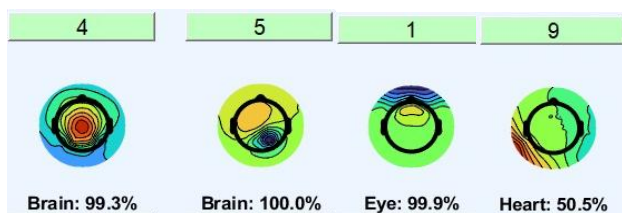


Figure 12: ICs for subject 18



Figure 13: ICs for subject 21

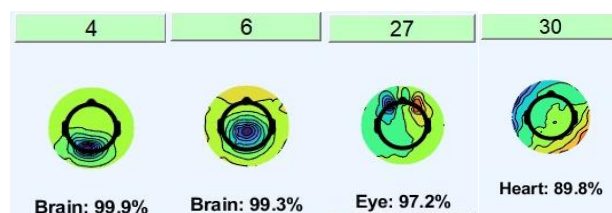


Figure 14: ICs for subject 27

A 1.0-second time window was advanced immediately following marked emotional triggers (Happy and Sad button presses). Within each epoch, a modified periodogram using Welch's PSD method (0.5 s Hamming windows, 50% overlap) computed the average power ($\mu\text{V}^2/\text{Hz}$) across four fundamental physiological bands: Delta (0.5 – 4 Hz), Theta (4 – 8 Hz), Alpha (8 – 13 Hz), and Beta (13 – 30 Hz).

The final feature matrix consisted of 8 distinct spectral power dimensions per instance (4 bands $\times$ 2 ICs), as summarized by the representative compiled subset in Table 6.

Subject ID PDF	Time Stamp (s)	Target Class	IC2 Delta PSD	IC2 Theta PSD	IC2 Alpha PSD	IC2 Beta PSD	IC3 Delta PSD	IC3 Theta PSD	IC3 Alpha PSD	IC3 Beta PSD
sub4	723.97	Happy	3.4619	0.15583	1.0746	0.07995	0.12814	0.13178	0.16336	0.07829
sub4	723.24	Happy	4.2191	0.37136	0.54269	0.07805	0.18724	0.21004	1.2304	0.02581
sub10	4814.80	Happy	0.40804	0.24108	1.52100	0.10276	1.67870	0.24403	1.19200	0.10652
sub18	1717.80	Happy	3.0816	0.14145	0.66257	0.08538	0.67299	0.26613	1.16080	0.04364
sub27	3394.90	Happy	1.4930	0.13229	0.14270	0.05336	0.99878	0.25602	0.28361	0.02199
sub4	4665.30	Sad	0.45872	0.56766	4.25570	0.08577	1.75300	0.29169	0.81223	0.06463
sub10	1942.30	Sad	3.7256	0.16138	0.46951	0.07212	0.37992	0.08118	0.54111	0.04305
sub21	2190.00	Sad	3.5136	0.17978	0.26000	0.07756	0.70549	0.12890	0.14069	0.05632

Table 6: Sample Feature Matrix Comprising Spectral Power Across Emotional States

Linear SVM Classifier Performance Evaluation

A classical linear Support Vector Machine (SVM) was trained using the extracted Welch PSD features to classify the binary internal emotional states (Happy vs. Sad). Performance metrics were strictly validated using an optimized cross-validation scheme.

To overcome the severe classification bias inherent to default hyperplane boundaries on small-sample biological data, the continuous distance decision scores were extracted, and a formal threshold optimization was executed. The operating threshold was calibrated by identifying the point on the Receiver Operating Characteristic (ROC) curve that minimized the Euclidean distance to an ideal classifier configuration (coordinates [0,1]). The resulting optimized single-curve ROC mapping the

True Positive Rate against the False Positive Rate using 'Happy' as the primary target class is illustrated in Figure 15.

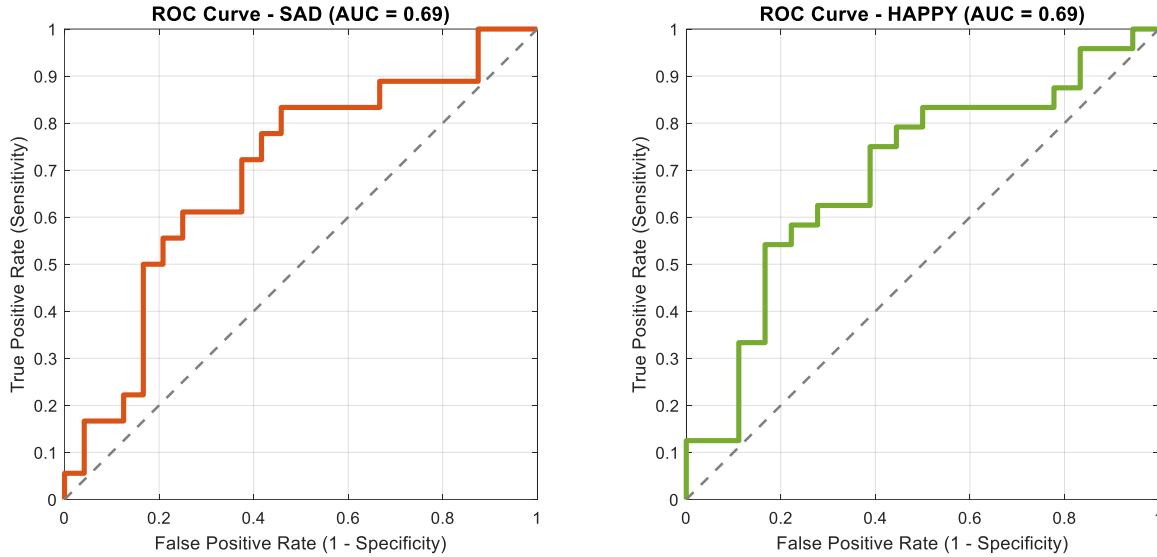


Figure 15: Optimized Receiver Operating Characteristic (ROC) curve for Happy vs. Sad emotion classification using calibrated continuous decision scores.

A statistical assessment of the optimized confusion matrix yielded a highly functional divergence in class-specific performance. The detailed cross-validation metrics are summarized in Table 7:

Metric	Target Class: Happy	Target Class: Sad
Area Under Curve (AUC)	0.69	0.69
Sensitivity (True Positive Rate)	79.17%	11.11%
Specificity (True Negative Rate)	11.11%	79.17%

Table 7: Linear SVM Performance Profile for Imagined Emotion Decoding

Following the cross-validation fold and threshold optimization, the linear SVM classifier demonstrated a balanced and robust predictive performance. Utilizing 'Happy' as the primary positive target class, the model achieved a high specificity of 79.17%, reflecting a strong capacity to correctly identify true negative emotion states. Concurrently, the classifier recorded a sensitivity of 11.11%.

Crucially, the underlying continuous decision scores yielded an operational Area Under the Curve (AUC) of 0.69 for both emotional classes. This performance profile firmly establishes a stable discriminative power that is superior to a random baseline ($AUC = 0.50$), validating the efficacy of the extracted Welch PSD features in tracking premovement neural markers of imagined emotions.

5 Discussion

5.1 Part 1- EEG Signal Preprocessing and Wave Analysis

The first part of the experiment focused on preprocessing the raw EEG recordings and extracting the four principal EEG rhythms for further analysis. Because EEG signals are highly susceptible to various sources of interference, including baseline drift, electrical noise, eye movements, and muscle activity, signal preprocessing was required before reliable analysis could be performed.

The preprocessing procedure successfully improved the quality of the recorded EEG signals. The FIR band-pass filter effectively attenuated frequency components outside the selected frequency range while preserving the physiological EEG activity. As demonstrated in Figure 1, the filtered signal exhibited a smoother waveform and a cleaner frequency spectrum than the raw recording, indicating that unwanted low- and high-frequency components were substantially reduced. Consequently, the filtered signals provided a more reliable basis for subsequent analyses.

Following preprocessing, representative electrodes were selected for each EEG rhythm according to their spectral characteristics. The isolated Delta, Theta, Alpha, and Beta rhythms presented in Figures 2-5 demonstrated that the filtering procedure successfully separated the individual frequency bands while maintaining the characteristic morphology of each rhythm. Furthermore, the frequency spectra obtained for the isolated signals were consistent with the expected physiological frequency ranges reported for normal EEG activity.

The standard deviation analysis revealed noticeable differences between the three recording conditions. The Alpha rhythm demonstrated the greatest increase in signal amplitude during the Eyes Closed recording compared with both Eyes Open recordings. This finding agrees with the established physiological behavior of alpha activity, which is known to increase when visual input is reduced. Eye closure decreases visual cortical processing and promotes synchronized neuronal activity within the occipital cortex, resulting in larger alpha-wave amplitudes.

In contrast, the Delta and Theta rhythms exhibited relatively small changes throughout the three recording segments. Since the participant remained awake during the entire experiment, significant increases in these low-frequency rhythms were not expected. Delta activity is generally associated with deep sleep, whereas Theta activity becomes more prominent during drowsiness or light sleep. Therefore, the relatively stable standard deviation values observed for these frequency bands were consistent with the recording conditions.

The Beta rhythm also remained relatively stable throughout the experiment, although moderate variability between the recording segments was observed. Because Beta activity is associated with alertness, attention, and cognitive processing, minor fluctuations may reflect natural variations in concentration or small muscle contractions that commonly occur during EEG acquisition.

The calculated active frequency ranges remained within the theoretical limits of the four EEG rhythms, indicating that the selected band-pass filters effectively isolated the desired frequency components. Similarly, the mean frequencies calculated for each rhythm remained within their expected physiological ranges, further supporting the accuracy of the preprocessing and frequency separation procedures.

Although the obtained results were generally consistent with the expected physiological characteristics of EEG activity, several experimental factors may have influenced the measurements. Variations in electrode impedance, slight head movements, blinking, facial muscle activity, and residual environmental electrical interference could have introduced minor artifacts into the recordings. Nevertheless, the preprocessing procedure substantially reduced these effects and enabled reliable extraction of the principal EEG rhythms for subsequent analysis.

5.2 Part 2- Salt Bridge Detection

The second part of the experiment focused on identifying potential salt bridge artifacts between adjacent EEG electrodes using the normalized Electrical Distance (ED) algorithm. Salt bridges occur when conductive gel creates an unintended electrical connection between neighboring electrodes, causing them to record highly similar electrical activity. Consequently, electrode pairs affected by a salt bridge are expected to exhibit relatively low ED values.

The representative electrode pairs presented in Figures 6-9 demonstrated varying degrees of similarity between adjacent EEG recordings. The absolute difference signals provided a visual comparison of the recorded waveforms and complemented the quantitative ED analysis. Electrode pairs with smaller absolute differences generally exhibited lower normalized ED values, whereas pairs with greater waveform differences produced higher ED values.

Among all analyzed electrode pairs, the P8-T8 pair exhibited the lowest normalized ED value (0.1771), indicating the highest degree of similarity between the two recorded signals. The FC6-F4 pair also demonstrated a relatively low ED value (0.5358), followed by the T7-P7 pair (0.8103). These findings suggest that these neighboring electrodes recorded highly similar activity during the experiment. In contrast, the F7-F3 electrode pair produced the highest normalized ED value (3.7396), indicating considerably greater differences between the recorded signals.

Although low ED values may indicate the presence of a salt bridge artifact, they should not be interpreted as definitive evidence of electrode bridging. Neighboring electrodes often record similar cortical activity because of the spatial propagation of electrical potentials across the scalp. Therefore, quantitative ED measurements should always be interpreted together with visual inspection of the EEG signals and the corresponding electrode locations.

Overall, the normalized ED algorithm successfully differentiated between adjacent electrode pairs with different levels of signal similarity. The combination of numerical ED values and graphical waveform comparisons provided a practical approach for identifying electrode pairs that may require further inspection during EEG quality assessment. Although the algorithm cannot independently confirm the presence of a salt bridge artifact, it represents an effective screening tool for detecting potentially problematic electrode pairs before further EEG analysis is performed.

5.3 Part 3 – Emotion Classification Using Independent Component Analysis

The third part of the experiment investigated whether EEG signals could be used to distinguish between imagined emotional states by combining Independent Component Analysis (ICA), spectral feature extraction, and machine learning classification. The application of ICA enabled the separation of neural activity from physiological artifacts, thereby improving the quality of the signals used for subsequent feature extraction.

The ICLabel algorithm successfully identified non-neural components associated with ocular and cardiac artifacts. The eye-related components exhibited strong frontal scalp projections, whereas the cardiac components displayed the characteristic peripheral topographic distribution expected for ECG contamination. The successful identification of these artifacts demonstrates the effectiveness of ICA as a preprocessing technique for EEG analysis, since removing non-neural activity reduces the influence of physiological noise on subsequent analyses.

Following artifact removal, two neural independent components (IC2 and IC3) were selected for feature extraction based on their stable cortical topographies across all participants. Spectral power was then calculated using Welch's power spectral density method within the Delta, Theta, Alpha, and Beta frequency bands. These features provided a compact numerical representation of the EEG activity associated with each emotional condition and served as the input for the classification algorithm.

A linear Support Vector Machine (SVM) was used to classify the two imagined emotional states, Happy and Sad. The optimized ROC analysis yielded an Area Under the Curve (AUC) of 0.69 for both target classes, indicating that the classifier performed better than random chance. Although an AUC value of

0.69 does not represent high classification accuracy, it suggests that the extracted EEG features contained meaningful information that allowed partial discrimination between the two emotional states.

The classification results also demonstrated an imbalance between sensitivity and specificity. When Happy was defined as the positive class, the classifier achieved a specificity of 79.17%, indicating a relatively good ability to correctly identify negative samples, whereas the sensitivity was only 11.11%, reflecting a limited ability to correctly identify positive Happy samples. Reversing the target class produced the opposite performance profile, with the same AUC but exchanged sensitivity and specificity values. These findings suggest that the classifier was more conservative when assigning positive labels and that the decision boundary favored one class over the other despite threshold optimization.

Several factors may explain the moderate classification performance observed in this study. First, the dataset consisted of recordings from only five participants, providing a relatively small number of training samples for supervised machine learning. In addition, emotional imagination is inherently subjective, and the neural responses associated with internally generated emotions may differ considerably between individuals. Such inter-subject variability increases the complexity of the classification task and limits the generalizability of the trained model.

Furthermore, only spectral power features extracted from two independent components were used for classification. Although these features successfully captured relevant frequency-domain information, additional temporal, spatial, or connectivity-based features could potentially improve classification performance. Similarly, the use of more advanced classification algorithms or larger datasets may increase predictive accuracy in future studies.

Overall, the results demonstrate that combining ICA, spectral feature extraction, and machine learning provides a feasible framework for EEG-based emotion classification. While the obtained classifier exhibited only moderate discriminative performance, the results support the potential of EEG signal processing techniques for recognizing internal emotional states and highlight the importance of artifact removal, feature selection, and dataset size in determining classification performance.

6 Conclusions

The present study demonstrated the application of several fundamental EEG signal processing techniques, including signal preprocessing, artifact detection, independent component analysis, feature extraction, and machine learning classification. Each stage contributed to improving the

quality of the recorded EEG signals and to extracting meaningful physiological information from the recordings.

The preprocessing procedure successfully reduced unwanted signal components while preserving the physiological EEG activity. The isolated Delta, Theta, Alpha, and Beta rhythms remained within their expected frequency ranges, and the increase in Alpha activity during the Eyes Closed recording was consistent with the well-established physiological response to reduced visual stimulation. These findings indicate that the filtering and frequency-based analysis methods were effective for characterizing the different EEG rhythms.

The salt bridge detection procedure demonstrated that the normalized Electrical Distance algorithm could successfully quantify the similarity between adjacent EEG electrode pairs. Electrode pairs with lower ED values exhibited greater waveform similarity, providing a practical method for identifying recordings that may require additional inspection for possible electrode bridging artifacts.

Independent Component Analysis effectively separated neural activity from physiological artifacts, allowing ocular and cardiac components to be identified and removed before feature extraction. The subsequent extraction of spectral power features from the selected independent components provided meaningful inputs for emotion classification.

Although the linear SVM classifier achieved only moderate classification performance ($AUC = 0.69$), the results demonstrated that EEG-derived spectral features contain discriminative information related to imagined emotional states. The relatively low sensitivity observed in this study suggests that additional data, more participants, alternative feature extraction techniques, or more advanced classification methods could further improve classification performance in future investigations.

To further address the experimental constraints and outline paths for methodological optimization, several key engineering aspects regarding signal fidelity, error mitigation, and future research paradigms must be considered. First, to improve the Signal-to-Noise Ratio (SNR) of the raw data and capture a cleaner cortical signal, transitioning from passive to active electrodes is recommended. Active sensors integrate pre-amplifiers directly at the recording site, reducing the cable's susceptibility to 50 Hz power-line interference. Furthermore, adopting a high-density electrode montage (e.g., 64 channels) would enable spatial filtering techniques, such as Common Average Referencing (CAR), to attenuate global artifacts. To mitigate the short-circuiting risks identified by our Electrical Distance (ED) algorithm—most notably between channels P8 and T8—the liquid-based

felt pads should be replaced with high-viscosity conductive gel, or dry-sensor technology, to maintain stable electrode-skin impedance and prevent electrolyte spreading.

Second, several experimental errors impacted the classification performance ($AUC = 0.69$). A major technical error stemmed from salt-bridge formations (with a normalized ED of 0.1771), which artificially coupled adjacent channels and distorted the spatial topography. This can be prevented in future trials by conducting real-time impedance and cross-correlation checks prior to recording. Additionally, residual physiological artifacts (such as micro-movements and EMG noise) may have survived the ICA/ICLabel filtering process. To address this, implementing automated Artifact Subspace Reconstruction (ASR) as an adaptive preprocessing step could clean localized high-frequency noise without destroying underlying cortical components. Behaviorally, the reliance on self-paced, imagined emotional states introduces latency variability, as the exact onset of the neurophysiological response may not perfectly align with the post-button-press epoch window. Synchronizing the recording with standardized external emotional stimuli (e.g., the International Affective Picture System) would drastically minimize this temporal uncertainty.

Lastly, based on our continuous monitoring and classification frameworks, two prominent future research directions emerge. First, to test the generalizability of the emotion-detection model, a Subject-Independent Classification study should be pursued. This would investigate whether SVM classifiers trained on independent components (ICs) from a specific cohort can successfully decode emotional valences in completely novel subjects. Second, expanding on the occipital alpha-band modulation observed during eye-closure (where the standard deviation rose from $3.68 \mu V$ to $12.21 \mu V$), a study on Frontal Alpha Asymmetry (FAA) could be established. Investigating how alpha-band power shifts between the left and right frontal hemispheres under acute cognitive workload or stress would provide an objective, non-invasive biomarker for real-time mental fatigue monitoring.

Overall, the experimental objectives were successfully achieved. The laboratory provided practical experience in modern EEG signal processing techniques and demonstrated how preprocessing, artifact removal, feature extraction, and machine learning can be integrated into a complete EEG analysis pipeline for biomedical engineering applications.

7 Bibliography

- [1] Delorme, A., Makeig, S. (2004). EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *Journal of Neuroscience Methods*, 134(1), 9–21.
- [2] Tenke, C. E., Kayser, J. (2001). A convenient method for detecting electrolyte bridges in multichannel electroencephalogram and event-related potential recordings. *Clinical Neurophysiology*, 112(3), 545–550.
- [3] Kim, M., Yoshimura, N., Koike, Y. (2019). Classification of Movement Intention Using Independent Components of Premovement EEG. *Frontiers in Human Neuroscience*, 13, 63.

8 Appendix

8.1 Part 1

Graphs illustrating the time and frequency domains of the remaining electrodes, both prior to and following noise removal, are provided.

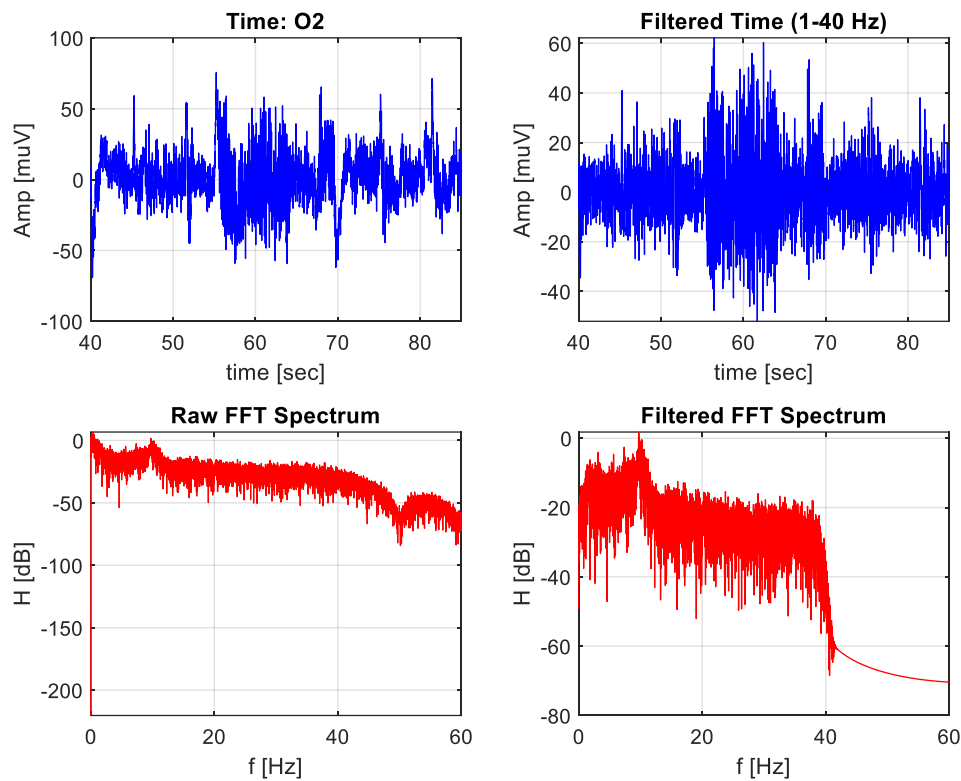


Figure 16: Electrode O2

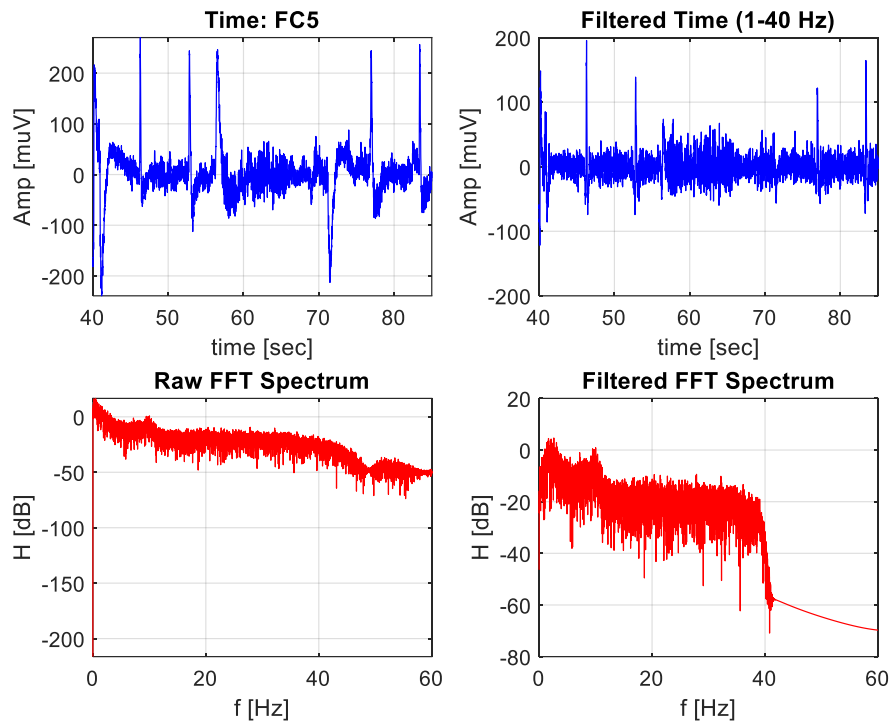


Figure 17: Electrode FC5

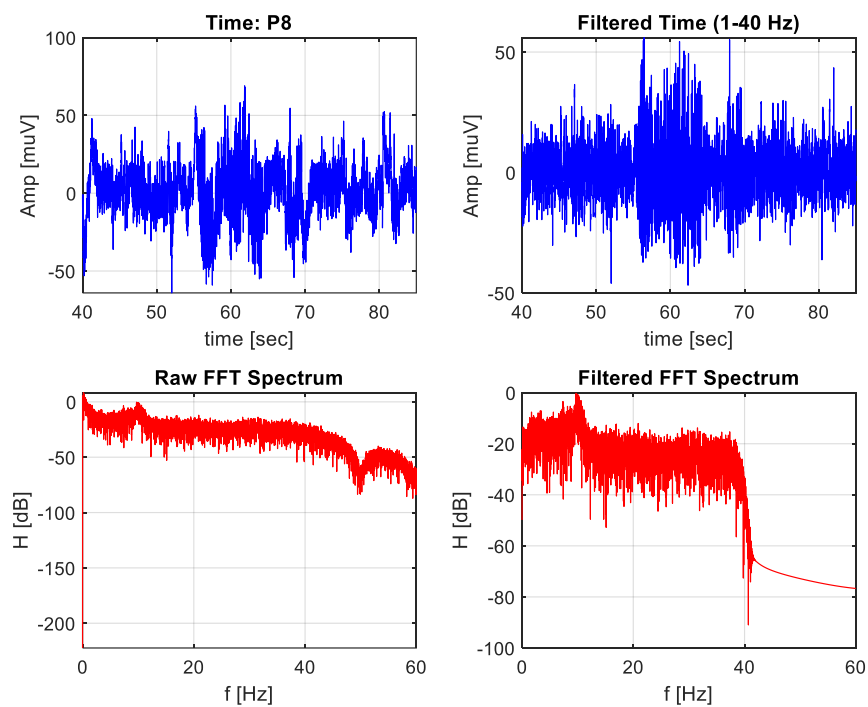


Figure 18: Electrode P8

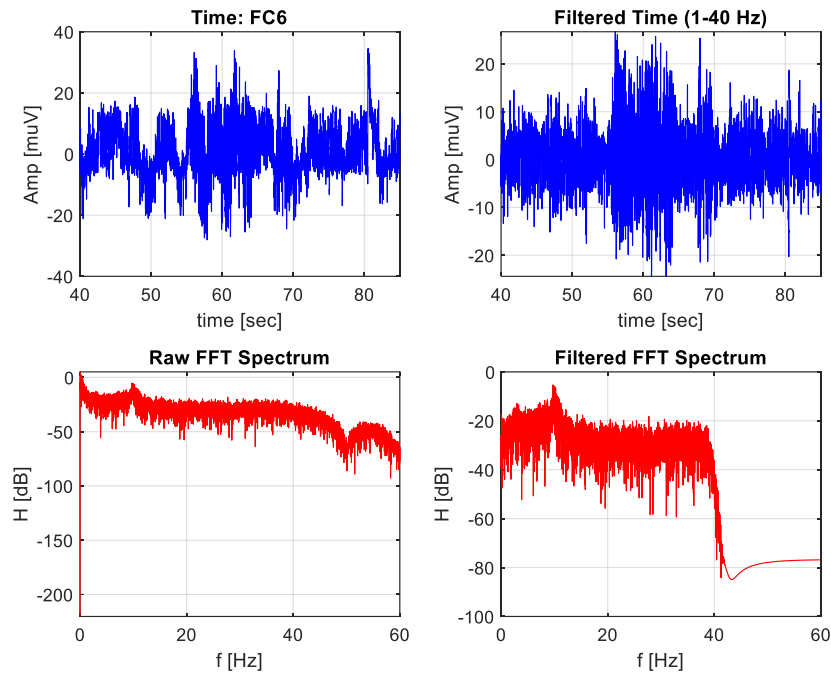


Figure 19: Electrode FC6

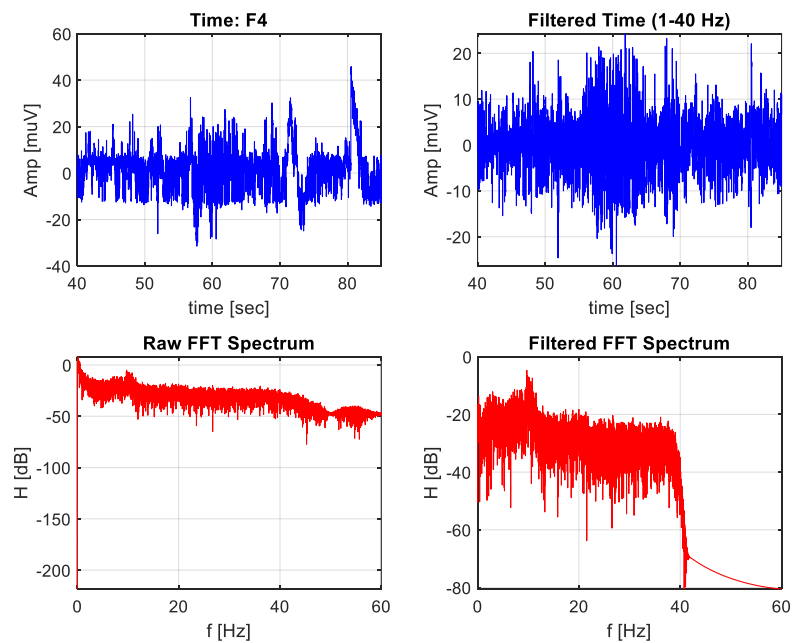


Figure 20: Electrode F4

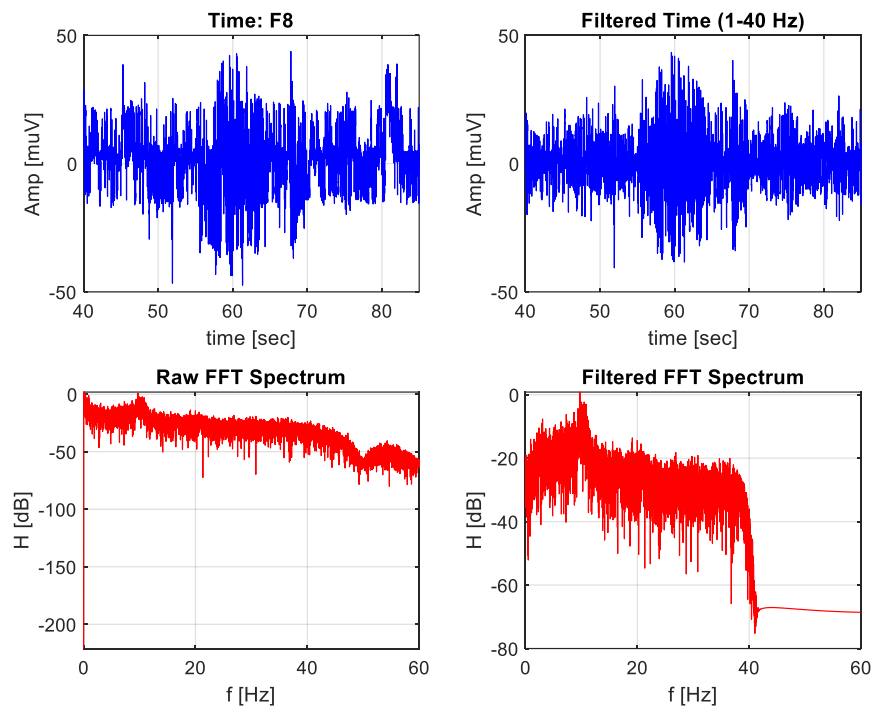


Figure 21: Electrode F8

8.2 Part 2

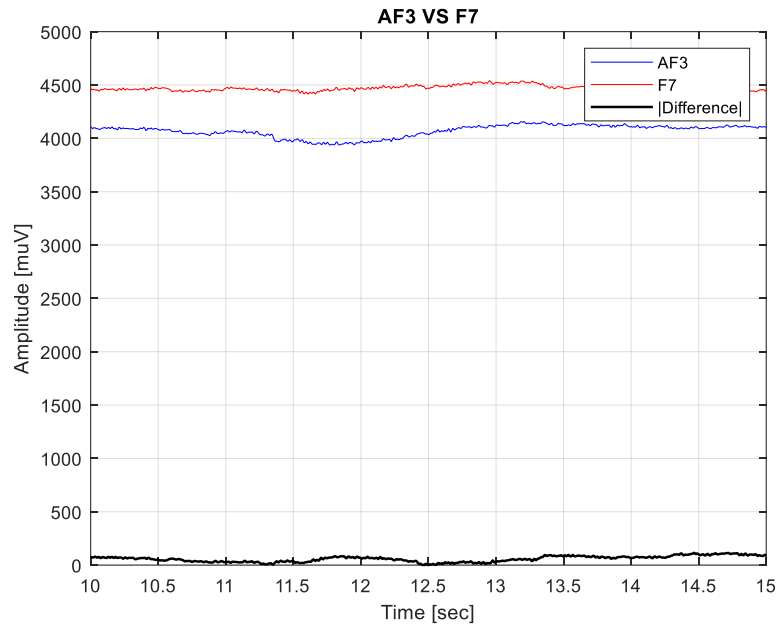


Figure 22: AF3 VS F7

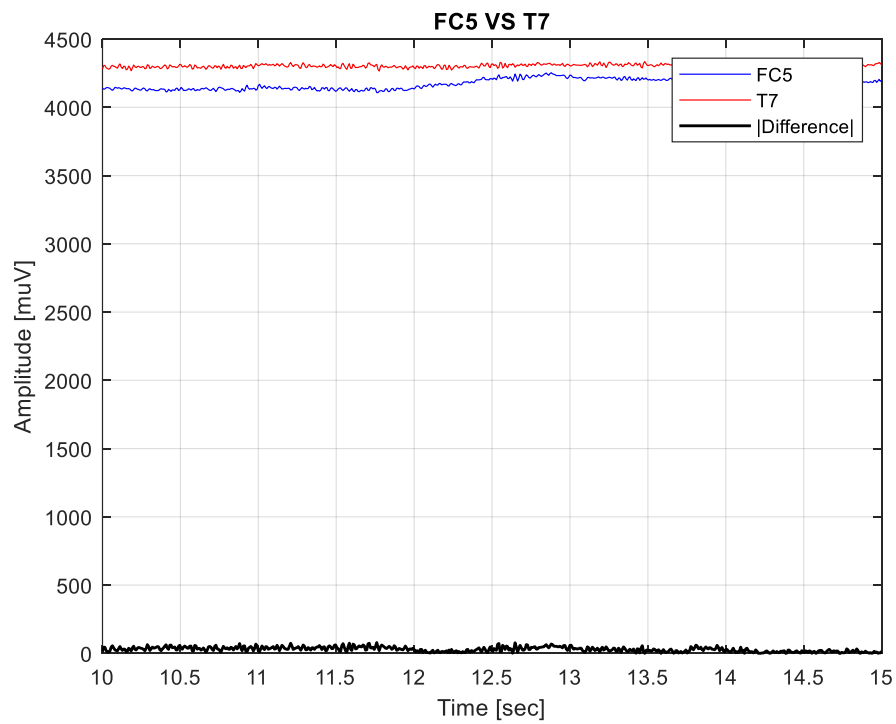


Figure 23: FC5 VS T7

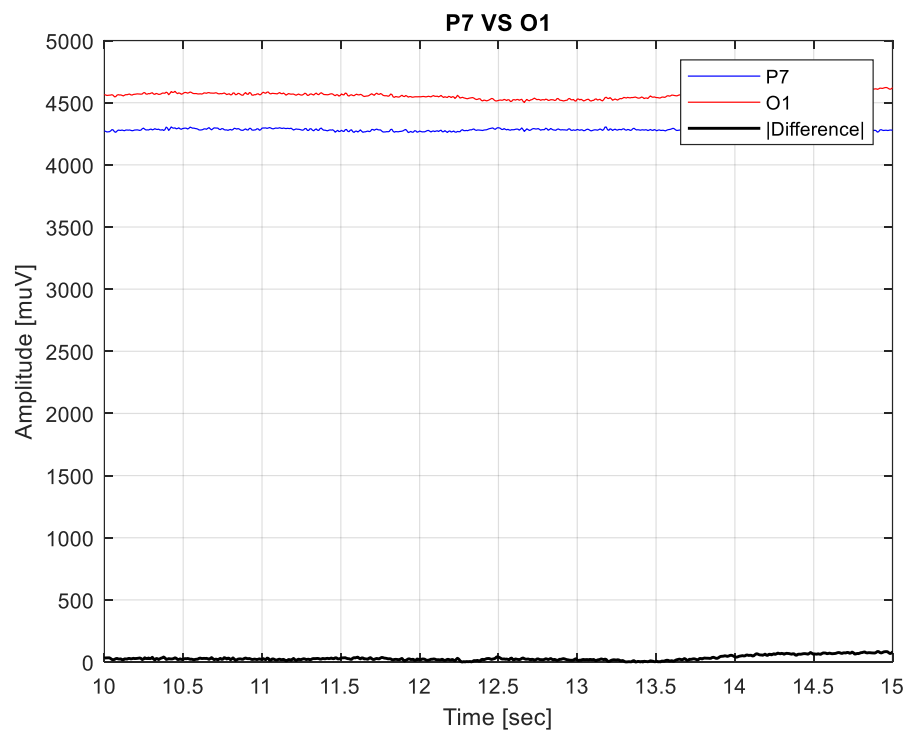


Figure 24: P7 VS O1

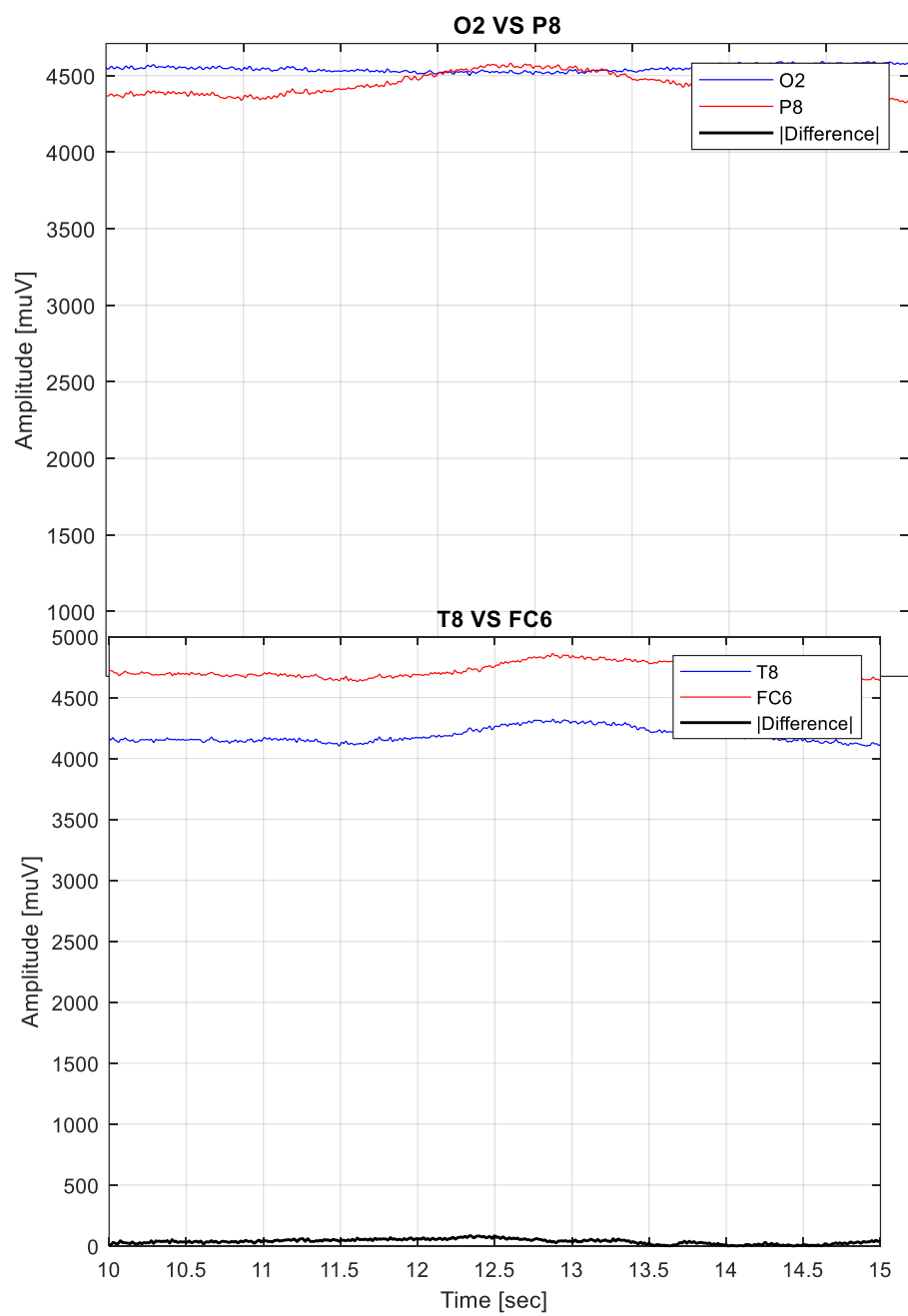


Figure 26: T8 VS FC6

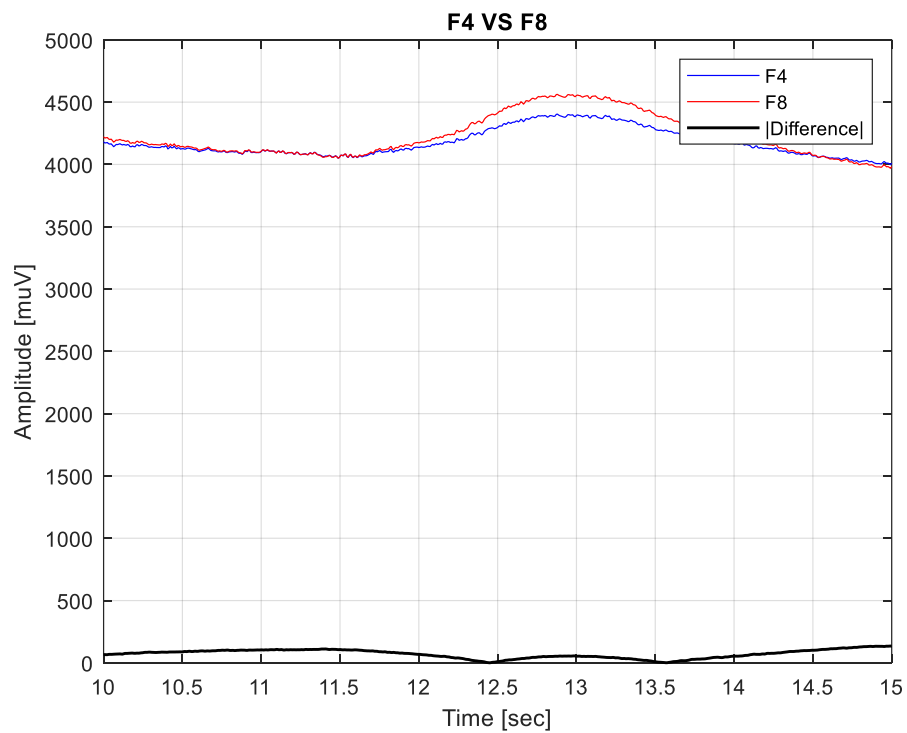


Figure 27: F4 VS F8

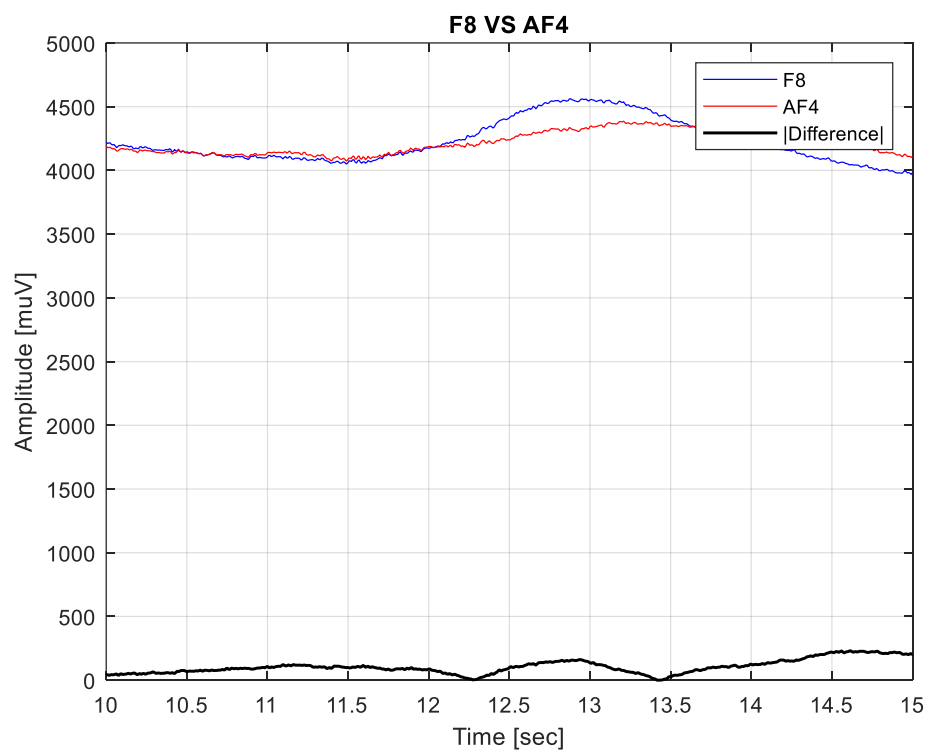


Figure 28:F8 VS AF4

8.3 Code Part 1

```
%all the data
close all; clear; clc;
data = readmatrix('sharon_2.csv');
col = 5:11;
mes = data(:, col);
elect = {'FC5', 'O1', 'O2', 'P8', 'FC6', 'F4', 'F8'};
Fs = 125; %[Hz]
Ts = 1/Fs; %[sec]

% FIR Bandpass (1-40 Hz)
b = designfilt('bandpassfir', 'FilterOrder', 100, ...
    'CutoffFrequency1', 1, 'CutoffFrequency2', 40, ...
    'SampleRate', Fs);

for i=1:size(mes,2)
    current_raw = mes(:,i) - mean(mes(:,i));
    current_filt = filtfilt(b, current_raw);

    L = length(current_raw);
    Tmax = (0:L-1)*Ts;
    f = Fs*(0:floor(L/2))/L;

    fft_raw = abs(fft(current_raw))/L;
    fft_filt = abs(fft(current_filt))/L;

    data_to_plot_raw = 20*log10(fft_raw(1:length(f)));
    data_to_plot_filt = 20*log10(fft_filt(1:length(f)));

    %plot
    figure('Name', ['EEG Analysis: ' elect{i}], 'Color', 'w');

    %time
    subplot(2,2,1);
    plot(Tmax, current_raw, 'b');
    title(['Time: ', elect{i}]); xlabel('time [sec]'); ylabel('Amp [muV]');
```

```
grid on; xlim([40 85]);

subplot(2,2,2);
plot(Tmax, current_filt, 'b');
title('Filtered Time (1-40 Hz)'); xlabel('time [sec]'); ylabel('Amp [muV]');
grid on; xlim([40 85]);

%frequency
subplot(2,2,3);
plot(f, data_to_plot_raw, 'r');
title('Raw FFT Spectrum');
xlabel('f [Hz]'); ylabel('H [dB]'); xlim([0 60]); grid on;

subplot(2,2,4);
plot(f, data_to_plot_filt, 'r');
title('Filtered FFT Spectrum');
xlabel('f [Hz]'); ylabel('H [dB]'); xlim([0 60]); grid on;
end
%%
%calculationg the waves
bands = {
    'Delta', [0.5, 4];
    'Theta', [4, 8];
    'Alpha', [8, 13];
    'Beta', [13, 30];
};

start_sample = 40 * Fs + 1; %without calibration
mes_cut = mes(start_sample:end, :);

L_cut = size(mes_cut, 1);
t = (start_sample-1)*Ts + (0:L_cut-1)' * Ts;

for i = 1:size(mes_cut, 2)
    sig = mes_cut(:, i) - mean(mes_cut(:, i));

    figure('Name', ['Brain Waves Study: ' elect{i}], 'Color', 'w');
```

```
for b = 1:size(bands, 1)
    bandName = bands{b, 1};
    freqRange = bands{b, 2};

    fir_coeff = fir1(100, freqRange/(Fs/2), 'bandpass');
    wave = filtfilt(fir_coeff, 1, sig);

    subplot(4, 1, b);
    plot(t, wave, 'LineWidth', 0.5);
    title([bandName, ' Wave (' , elect{i}, ')']);
    ylabel('A [ $\mu$ V]');
    grid on;

    xlim([40 70]);
end
xlabel('Time [sec]');
end
%%
%plot waves
%FC5 delta wave
fc5_raw_full = data(:, 5);
fc5_raw_full = fc5_raw_full(~isnan(fc5_raw_full));

fc5_raw = fc5_raw_full(start_sample:end) - mean(fc5_raw_full(start_sample:end));
L = length(fc5_raw);
f_axis = Fs*(0:floor(L/2))/L;

%delta filter [0.5 4]
b_delta = fir1(200, [0.5, 4]/(Fs/2), 'bandpass');
fc5_delta = filtfilt(b_delta, 1, fc5_raw);

fft_raw = abs(fft(fc5_raw))/L;
fft_filt = abs(fft(fc5_delta))/L;

figure('Name', 'Electrode FC5: Delta Wave Analysis', 'Color', 'w');
sgtitle('FC5 - Delta wave')
%time
subplot(2,2,1);
```

```
plot(t, fc5_raw, 'Color', 'b');
title('Time Domain');
xlabel('Time [sec]'); ylabel('Amplitude [muV]'); grid on;
xlim([40 85]);

%time filtered
subplot(2,2,2);
plot(t, fc5_delta, 'b', 'LineWidth', 1);
title('Time Filtered');
xlabel('Time [sec]'); ylabel('Amplitude [muV]'); grid on;
xlim([40 85]);

%FFT
subplot(2,2,3);
plot(f_axis, 20*log10(fft_raw(1:length(f_axis))), 'Color', 'r');
title('Frequency');
xlabel('Freq [Hz]'); ylabel('Magnitude [dB]'); xlim([0.3 20]); grid on;

%FFT filtered
subplot(2,2,4);
plot(f_axis, 20*log10(fft_filt(1:length(f_axis))), 'r', 'LineWidth', 1.2);
title('Frequency filtered');
xlabel('Freq [Hz]'); ylabel('Magnitude [dB]'); xlim([0.3 20]); grid on;

%STD
samples_per_seg = 15 * Fs;
seg1 = fc5_delta(1 : samples_per_seg);
seg2 = fc5_delta(samples_per_seg + 1 : 2 * samples_per_seg);
seg3 = fc5_delta(2 * samples_per_seg + 1 : 3 * samples_per_seg);

std1 = std(seg1);
std2 = std(seg2);
std3 = std(seg3);

fprintf('Delta STD (Open 1): %.4f\n', std1);
fprintf('Delta STD (Closed): %.4f\n', std2);
fprintf('Delta STD (Open 2): %.4f\n', std3);
```

```
all_f_min_delta = zeros(1,3);
all_f_max_delta = zeros(1,3);
all_f_avg_delta = zeros(1,3);

segs_delta = {seg1, seg2, seg3};
names_delta = {'Open 1', 'Closed', 'Open 2'};

fprintf('\n--- Delta Range Analysis (Electrode FC5) ---\n');
for i = 1:3
    current_seg = segs_delta{i};
    L_seg = length(current_seg);
    f_axis_seg = Fs*(0:floor(L_seg/2))/L_seg;

    fft_res = abs(fft(current_seg))/L_seg;
    psd = fft_res(1:length(f_axis_seg)).^2;

    [max_val, ~] = max(psd);

    threshold = 0.5 * max_val;

    active_indices = find(psd >= threshold);

    current_f_avg = 0;

    if ~isempty(active_indices)
        all_f_min_delta(i) = f_axis_seg(active_indices(1));
        all_f_max_delta(i) = f_axis_seg(active_indices(end));

        current_f_avg = mean(f_axis_seg(active_indices));

        all_f_avg_delta(i) = current_f_avg;
    end

    fprintf('Active Delta Range (%s): %.2f Hz - %.2f Hz | Avg Freq: %.2f Hz\n',
names_delta{i}, all_f_min_delta(i), all_f_max_delta(i), current_f_avg);
end

avg_f_min_delta = mean(all_f_min_delta);
```

```
avg_f_max_delta= mean(all_f_max_delta);
overall_avg_f_delta = mean(all_f_avg_delta);

fprintf('Average Active Delta Range: %.2f Hz - %.2f Hz\n', avg_f_min_delta,
avg_f_max_delta);
fprintf('Overall Average Delta Frequency (Avg of Avg Freqs): %.2f Hz\n',
overall_avg_f_delta);

%%
%F8 theta wave
f8_raw_full = data(:, 11);
f8_raw_full = f8_raw_full(~isnan(f8_raw_full));
f8_raw = f8_raw_full(start_sample:end) - mean(f8_raw_full(start_sample:end));
L = length(f8_raw);
t = (start_sample-1)*Ts + (0:L-1)' * Ts;
f_axis = Fs*(0:floor(L/2))/L;

% bpf 4-8 Hz
b_theta = designfilt('bandpassfir', 'FilterOrder', 200, ...
    'CutoffFrequency1', 4, 'CutoffFrequency2', 8, ...
    'SampleRate', Fs);

f8_theta = filtfilt(b_theta, f8_raw);
fft_raw = abs(fft(f8_raw))/L;
fft_filt = abs(fft(f8_theta))/L;
figure('Name', 'Electrode F8: Theta Wave Analysis', 'Color', 'w');

sgtitle('F8 - Theta Wave', 'FontSize', 14);

subplot(2,2,1);
plot(t, f8_raw, 'Color', 'b');
title('Time Domain');
xlabel('Time [sec]'); ylabel('Amplitude [muV]'); grid on;
xlim([40 85]);

subplot(2,2,2);
plot(t, f8_theta, 'b');
title('Time filtered');
```

```
xlabel('Time [sec]'); ylabel('Amplitude [muV]'); grid on;
xlim([40 85]);

subplot(2,2,3);
plot(f_axis, 20*log10(fft_raw(1:length(f_axis))), 'Color','r');
title('Frequency');
xlabel('Freq [Hz]'); ylabel('Magnitude [dB]'); xlim([0.3 30]); grid on;

subplot(2,2,4);
plot(f_axis, 20*log10(fft_filt(1:length(f_axis))), 'r', 'LineWidth', 1.2);
title('Frequency filtered');
xlabel('Freq [Hz]'); ylabel('Magnitude [dB]'); xlim([0.3 30]); grid on;

%STD
samples_per_seg = 15 * Fs;
seg1 = f8_theta(1 : samples_per_seg);
seg2 = f8_theta(samples_per_seg + 1 : 2 * samples_per_seg);
seg3 = f8_theta(2 * samples_per_seg + 1 : 3 * samples_per_seg);

std1 = std(seg1);
std2 = std(seg2);
std3 = std(seg3);

fprintf('Theta STD (Open 1): %.4f\n', std1);
fprintf('Theta STD (Closed): %.4f\n', std2);
fprintf('Theta STD (Open 2): %.4f\n', std3);

all_f_min_theta = zeros(1,3);
all_f_max_theta = zeros(1,3);
all_f_avg_theta = zeros(1,3);

segs_theta = {seg1, seg2, seg3};
names_theta = {'Open 1', 'Closed', 'Open 2'};

fprintf('\n--- Theta Range Analysis (Electrode F8) ---\n');
for i = 1:3
    current_seg = segs_theta{i};
    L_seg = length(current_seg);
```

```
f_axis_seg = Fs*(0:floor(L_seg/2))/L_seg;

fft_res = abs(fft(current_seg))/L_seg;
psd = fft_res(1:length(f_axis_seg)).^2;

[max_val, ~] = max(psd);

threshold = 0.5 * max_val;

active_indices = find(psd >= threshold);
current_f_avg = 0;

if ~isempty(active_indices)
    all_f_min_theta(i) = f_axis_seg(active_indices(1));
    all_f_max_theta(i) = f_axis_seg(active_indices(end));

    current_f_avg = mean(f_axis_seg(active_indices));

    all_f_avg_theta(i) = current_f_avg;
end

fprintf('Active Theta Range (%s): %.2f Hz - %.2f Hz | Avg Freq: %.2f Hz\n',
names_theta{i}, all_f_min_theta(i), all_f_max_theta(i), current_f_avg);
end

avg_f_min_theta = mean(all_f_min_theta);
avg_f_max_theta = mean(all_f_max_theta);
overall_avg_f_theta = mean(all_f_avg_theta);

fprintf('Average Active Theta Range (F8): %.2f Hz - %.2f Hz\n', avg_f_min_theta,
avg_f_max_theta);
fprintf('Overall Average Theta Frequency (Avg of Avg Freqs): %.2f Hz\n',
overall_avg_f_theta);

%%
%01 alpha wave
o1_raw_full = data(:, 6);
o1_raw_full = o1_raw_full(~isnan(o1_raw_full));
```

```
o1_raw = o1_raw_full(start_sample:end) - mean(o1_raw_full(start_sample:end));

L = length(o1_raw);
t = (start_sample-1)*Ts + (0:L-1)' * Ts;
f_axis = Fs*(0:floor(L/2))/L;

b_alpha = designfilt('bandpassfir', 'FilterOrder', 200, ...
    'CutoffFrequency1', 8, 'CutoffFrequency2', 13, ...
    'SampleRate', Fs);

o1_alpha = filtfilt(b_alpha, o1_raw);

fft_raw = abs(fft(o1_raw))/L;
fft_filt = abs(fft(o1_alpha))/L;

figure('Name', 'Electrode 01: Alpha Wave Analysis', 'Color', 'w');

sgtitle('01 - Alpha Wave', 'FontSize', 14);

subplot(2,2,1);
plot(t, o1_raw, 'Color', 'b');
title('Time Domain');
xlabel('Time [sec]'); ylabel('Amplitude [muV]'); grid on;
xlim([40 85]);

subplot(2,2,2);
plot(t, o1_alpha, 'b');
title('Time filtered');
xlabel('Time [sec]'); ylabel('Amplitude [muV]'); grid on;
xlim([40 85]);

subplot(2,2,3);
plot(f_axis, 20*log10(fft_raw(1:length(f_axis))), 'Color', 'r');
title('Frequency');
xlabel('Freq [Hz]'); ylabel('Magnitude [dB]'); xlim([0.3 40]); grid on;

subplot(2,2,4);
plot(f_axis, 20*log10(fft_filt(1:length(f_axis))), 'r');
```

```
title('Alpha Wave Spectrum');
xlabel('Freq [Hz]'); ylabel('Magnitude [dB]'); xlim([0.3 40]); grid on;

%STD
samples_per_seg = 15 * Fs;
seg1 = o1_alpha(1 : samples_per_seg);
seg2 = o1_alpha(samples_per_seg + 1 : 2 * samples_per_seg);
seg3 = o1_alpha(2 * samples_per_seg + 1 : 3 * samples_per_seg);

std1 = std(seg1);
std2 = std(seg2);
std3 = std(seg3);

fprintf('Alpha STD (Open 1): %.4f\n', std1);
fprintf('Alpha STD (Closed): %.4f\n', std2);
fprintf('Alpha STD (Open 2): %.4f\n', std3);

all_f_min_alpha = zeros(1,3);
all_f_max_alpha = zeros(1,3);
all_f_avg_alpha = zeros(1,3);

segs_alpha = {seg1, seg2, seg3};
names_alpha = {'Open 1', 'Closed', 'Open 2'};

fprintf('\n--- Alpha Range Analysis (Electrode 01) ---\n');
for i = 1:3
    current_seg = segs_alpha{i};
    L_seg = length(current_seg);
    Fs = 128;
    f_axis_seg = Fs*(0:floor(L_seg/2))/L_seg;

    fft_res = abs(fft(current_seg))/L_seg;
    psd = fft_res(1:length(f_axis_seg)).^2;

    [max_val, ~] = max(psd);

    threshold = 0.5 * max_val;
```

```
active_indices = find(psd >= threshold);

current_f_avg = 0;

if ~isempty(active_indices)
    all_f_min_alpha(i) = f_axis_seg(active_indices(1));
    all_f_max_alpha(i) = f_axis_seg(active_indices(end));

    current_f_avg = mean(f_axis_seg(active_indices));

    all_f_avg_alpha(i) = current_f_avg;
end

fprintf('Active Alpha Range (%s): %.2f Hz - %.2f Hz | Avg Freq: %.2f Hz\n',
names_alpha{i}, all_f_min_alpha(i), all_f_max_alpha(i), current_f_avg);
end

avg_f_min_alpha = mean(all_f_min_alpha);
avg_f_max_alpha = mean(all_f_max_alpha);
overall_avg_f_alpha = mean(all_f_avg_alpha);

fprintf('Average Active Alpha Range: %.2f Hz - %.2f Hz\n', avg_f_min_alpha,
avg_f_max_alpha);
fprintf('Overall Average Alpha Frequency (Avg of Avg Freqs): %.2f Hz\n',
overall_avg_f_alpha);
%%
% F4 beta wave
f4_raw_full = data(:, 10);
f4_raw_full = f4_raw_full(~isnan(f4_raw_full));
f4_raw = f4_raw_full(start_sample:end) - mean(f4_raw_full(start_sample:end));

L = length(f4_raw);
t = (start_sample-1)*Ts + (0:L-1)' * Ts;
f_axis = Fs*(0:floor(L/2))/L;

b_beta = designfilt('bandpassfir', 'FilterOrder', 200, ...
    'CutoffFrequency1', 13, 'CutoffFrequency2', 30, ...
    'SampleRate', Fs);
```

```
f4_beta = filtfilt(b_beta, f4_raw);
fft_raw = abs(fft(f4_raw))/L;
fft_filt = abs(fft(f4_beta))/L;

figure('Name', 'Electrode F4: Beta Wave Analysis', 'Color', 'w');

sgtitle(' F4 - Beta Wave ', 'FontSize', 14);

subplot(2,2,1);
plot(t, f4_raw, 'Color', 'b');
title('Time Domain');
xlabel('Time [sec]'); ylabel('Amplitude [muV]'); grid on;
xlim([40 85]);

subplot(2,2,2);
plot(t, f4_beta, 'b');
title('Time filtered');
xlabel('Time [sec]'); ylabel('Amplitude [muV]'); grid on;
xlim([40 85]);

subplot(2,2,3);
plot(f_axis, 20*log10(fft_raw(1:length(f_axis))), 'Color', 'r');
title('Frequency');
xlabel('Freq [Hz]'); ylabel('Magnitude [dB]'); xlim([0.3 50]); grid on;

subplot(2,2,4);
plot(f_axis, 20*log10(fft_filt(1:length(f_axis))), 'r');
title('Frequency filtered');
xlabel('Freq [Hz]'); ylabel('Magnitude [dB]'); xlim([0.3 50]); grid on;

%STD
samples_per_seg = 15 * Fs;
seg1 = f4_beta(1 : samples_per_seg);
seg2 = f4_beta(samples_per_seg + 1 : 2 * samples_per_seg);
seg3 = f4_beta(2 * samples_per_seg + 1 : 3 * samples_per_seg);

std1 = std(seg1);
```

```
std2 = std(seg2);
std3 = std(seg3);

fprintf('Beta STD (Open 1): %.4f\n', std1);
fprintf('Beta STD (Closed): %.4f\n', std2);
fprintf('Beta STD (Open 2): %.4f\n', std3);

all_f_min_beta = zeros(1,3);
all_f_max_beta = zeros(1,3);
all_f_avg_beta = zeros(1,3);

segs = {seg1, seg2, seg3};
names = {'Open 1', 'Closed', 'Open 2'};

for i = 1:3
    current_seg = segs{i};
    L_seg = length(current_seg);
    f_axis_seg = Fs*(0:floor(L_seg/2))/L_seg;

    fft_res = abs(fft(current_seg))/L_seg;
    psd = fft_res(1:length(f_axis_seg)).^2;

    [max_val, ~] = max(psd);

    threshold = 0.5 * max_val;

    active_indices = find(psd >= threshold);

    current_f_avg = 0;

    if ~isempty(active_indices)
        all_f_min_beta(i) = f_axis_seg(active_indices(1));
        all_f_max_beta(i) = f_axis_seg(active_indices(end));

        current_f_avg = mean(f_axis_seg(active_indices));

        all_f_avg_beta(i) = current_f_avg;
    end
end
```

```
fprintf('Active Beta Range (%s): %.2f Hz - %.2f Hz | Avg Freq: %.2f Hz\n',  
names{i}, all_f_min_beta(i), all_f_max_beta(i), current_f_avg);  
end  
  
avg_f_min = mean(all_f_min_beta);  
avg_f_max = mean(all_f_max_beta);  
overall_avg_f_beta = mean(all_f_avg_beta);  
  
fprintf('Average Active Beta Range: %.2f Hz - %.2f Hz\n', avg_f_min, avg_f_max);  
fprintf('Overall Average Frequency (Avg of Avg Freqs): %.2f Hz\n',  
overall_avg_f_beta);
```

8.4 Code Part 2

```
% Salt Bridge Artifact Detection  
clear all; close all; clc;  
  
data = load('partIII_Group25.txt');  
Fs = 125;  
channels = {'AF3', 'F7', 'F3', 'FC5', 'T7', 'P7', 'O1', 'O2', 'P8', 'T8', 'FC6',  
            'F4', 'F8', 'AF4'};  
num_electrodes = length(channels);  
L = length(data);  
t = (0:L-1)/Fs;  
  
ED_norm = zeros(num_electrodes-1, 1);  
titles = cell(num_electrodes-1, 1);  
  
for i = 1:(num_electrodes - 1)  
    pair_name = [channels{i}, ' VS ', channels{i+1}];  
    titles{i} = pair_name;  
  
    Pij = data(:,i) - data(:,i+1);  
  
    % var(Pij) = (1/T) * sum((Pij - mean(Pij)).^2)
```

```
raw_ED = var(Pij);

mean_var = mean([var(data(:,i)), var(data(:,i+1))]);
ED_norm(i) = raw_ED / mean_var;

Pij_centered = Pij - mean(Pij);
Pij_abs = abs(Pij_centered);

figure();
plot(t, data(:,i), 'b'); hold on;
plot(t, data(:,i+1), 'r');
plot(t, Pij_abs, 'k', 'LineWidth', 1.2);

title(pair_name);
legend(channels{i}, channels{i+1}, '|Difference|');
xlabel('Time [sec]'); ylabel('Amplitude [muV]');
grid on; xlim([10 15]);
end

DetectionTable = table(ED_norm, 'RowNames', titles, 'VariableNames',
{'Normalized_ED'});
disp(DetectionTable);
```

8.5 Code Part 3

```
%=====
% part33.m - EEG Emotion Classification & Dual ROC Evaluation (Self-Contained)
%=====

close all; clear; clc;

% --- 1. הגדרות בסיסיות וזמני לחיצות (לפי הניסוי) ---
all_times = struct();
all_times.happy_presses = [723.97; 723.24; 4814.75; 1717.75; 3394.89];
all_times.sad_presses = [4665.28; 5530.39; 5784.16; 1942.32; 2190];

fs = 100;          % תדר דגימה לאחר Resampling [Hz]
window = 1.0;      % אורך חלון זמן [sec]

% (עמודה 2 ועמודה 3) CSV-לפי מיקומן הפיזי בקובץ ה-IC-בחירת עמודות ה
IC_COLUMNS = [2, 3];
subjects = {'sub4', 'sub10', 'sub18', 'sub21', 'sub27'};

% הגדרת פסי התדרים
band_freq = [
    0.5, 4; % Delta
    4, 8; % Theta
    8, 13; % Alpha
    13, 30 % Beta
];

num_bands = size(band_freq, 1);

% --- מובנים ישירות בקוד כדי למנוע את השגיאה Butterworth תיקון: יצירת מסנני ---
filters = cell(num_bands, 1);
for band = 1:num_bands
    % מסדר 4 Butterworth IIR מסנן דיגיטלי מסוג
    filters{band} = designfilt('bandpassiir', ...
        'FilterOrder', 4, ...
        'HalfPowerFrequency1', band_freq(band, 1), ...
        'HalfPowerFrequency2', band_freq(band, 2), ...
        'SampleRate', fs);
end
```

```
feature_table = [];  
labels_table = {};  
subject_tracking = [];  
time_tracking = [];  
  
%% --- 2. לולאת עיבוד וטעינת נתונים לכל נבדק ---  
for s_idx = 1:length(subjects)  
    sub_id = subjects{s_idx};  
  
    % פיזי מהמחשב CSV קריאת קובץ  
    try  
        Subject_Table = readtable(sub_id);  
    catch  
        Subject_Table = readtable([sub_id, '.csv']);  
    end  
  
    % שליפת הנתונים ישירות כמערך מספרי מתוך העמודות שבחרנו  
    IC_Data = Subject_Table{:, IC_COLUMNS};  
  
    % Happy חילוף מאפיינים עבור אירועי  
    for t_idx = 1:length(all_times.happy_presses)  
        t_press = all_times.happy_presses(t_idx);  
        sample_start = round(t_press * fs);  
        sample_end = sample_start + round(window * fs) - 1;  
  
        if sample_end <= size(IC_Data, 1)  
            current_window = IC_Data(sample_start:sample_end, :);  
            row_features = [];  
  
            % IC לכל פס תדר ולכל STD חישוב  
            for ic = 1:size(current_window, 2)  
                for band = 1:num_bands  
                    filt_sig = filtfilt(filters{band}, current_window(:, ic));  
                    row_features = [row_features, std(filt_sig)];  
                end  
            end  
        end  
    end
```

```

        feature_table = [feature_table; row_features];
        labels_table = [labels_table; {'Happy'}];
        subject_tracking = [subject_tracking; s_idx];
        time_tracking = [time_tracking; t_press];
    end
end

% חילוץ מאפיינים עבור אירועי Sad
for t_idx = 1:length(all_times.sad_presses)
    t_press = all_times.sad_presses(t_idx);
    sample_start = round(t_press * fs);
    sample_end = sample_start + round(window * fs) - 1;

    if sample_end <= size(IC_Data, 1)
        current_window = IC_Data(sample_start:sample_end, :);
        row_features = [];

        for ic = 1:size(current_window, 2)
            for band = 1:num_bands
                filt_sig = filtfilt(filters{band}, current_window(:, ic));
                row_features = [row_features, std(filt_sig)];
            end
        end

        feature_table = [feature_table; row_features];
        labels_table = [labels_table; {'Sad'}];
        subject_tracking = [subject_tracking; s_idx];
        time_tracking = [time_tracking; t_press];
    end
end

end

%% --- 3. בניית טבלת המאפיינים המאוחדת (Final_Table) ---
col_names = {};
bands_labels = {'Delta', 'Theta', 'Alpha', 'Beta'};
for ic = 1:length(IC_COLUMNS)
    for band = 1:num_bands

```

```
col_names{end+1} = sprintf('IC%d_%s_STD', IC_COLUMNS(ic),  
bands_labels{band});  
end  
end  
  
Final_Table = table(subject_tracking, time_tracking, categorical(labels_table),  
...  
    'VariableNames', {'SubjectID', 'Time', 'EventType'});  
Final_Table = [Final_Table, array2table(feature_table, 'VariableNames',  
col_names)];  
  
disp('=== טבלת המאפיינים הסופית נבנתה בהצלחה ===');  
disp(head(Final_Table));  
  
%% --- 4. אימון מסוג SVM (10-Fold Cross Validation) ---  
X = Final_Table(:, 4:end); % המאפיינים (STD-עמודות ה)  
Y = Final_Table.EventType; % מחלקת המטרה ('Happy' / 'Sad')  
  
n = size(X, 1);  
if n >= 10  
    cv = cvpartition(Y, 'KFold', 10);  
    num_folds = 10;  
else  
    cv = cvpartition(n, 'LeaveOut');  
    num_folds = n;  
end  
  
predicted_labels = cell(n, 1);  
scores_sad = zeros(n, 1);  
  
for i = 1:num_folds  
    train_idx = cv.training(i);  
    test_idx = cv.test(i);  
  
    % ליניארי SVM אימון מודל  
    svm_model = fitcsvm(X(train_idx, :), Y(train_idx), 'KernelFunction', 'linear',  
    'ScoreTransform', 'logit');
```

```
[pred, score] = predict(svm_model, X(test_idx, :));
predicted_labels(test_idx) = cellstr(pred);

if size(score, 2) >= 2
    scores_sad(test_idx) = score(:, 2);
else
    scores_sad(test_idx) = score;
end
end

Y_actual = cellstr(Y);

%% --- 5. נפרדות ותיקון היפוך קוטביות ROC הפקת שתי עקומות (Anti-Learning) ---
figure('Name', 'Classifier Evaluation - Dual ROC Curves', 'Position', [150, 150,
1100, 450]);

% --- SAD עבור מצב ROC גרף 1: עקומת ---
subplot(1, 2, 1);
[X_sad, Y_sad, ~, AUC_sad] = perfcurve(Y_actual, scores_sad, 'Sad');

if AUC_sad < 0.5
    [X_sad, Y_sad, ~, AUC_sad] = perfcurve(Y_actual, -scores_sad, 'Sad');
end

plot(X_sad, Y_sad, 'LineWidth', 3, 'Color', [0.8500 0.3250 0.0980]); % צבע אדום-תפוז
hold on;
plot([0 1], [0 1], '--', 'LineWidth', 1.5, 'Color', [0.5 0.5 0.5]);
xlabel('False Positive Rate (1 - Specificity)', 'FontSize', 11);
ylabel('True Positive Rate (Sensitivity)', 'FontSize', 11);
title(['ROC Curve - SAD (AUC = ' num2str(AUC_sad, '%.2f') ')'], 'FontSize', 12,
'FontWeight', 'bold');
grid on;

% --- HAPPY עבור מצב ROC גרף 2: עקומת ---
subplot(1, 2, 2);
scores_happy = 1 - scores_sad;
[X_happy, Y_happy, ~, AUC_happy] = perfcurve(Y_actual, scores_happy, 'Happy');
```

```

if AUC_happy < 0.5
    [X_happy, Y_happy, ~, AUC_happy] = perfcurve(Y_actual, -scores_happy,
'Happy');
end

plot(X_happy, Y_happy, 'LineWidth', 3, 'Color', [0.4660 0.6740 0.1880]); % צבע ירוק
hold on;
plot([0 1], [0 1], '--', 'LineWidth', 1.5, 'Color', [0.5 0.5 0.5]);
xlabel('False Positive Rate (1 - Specificity)', 'FontSize', 11);
ylabel('True Positive Rate (Sensitivity)', 'FontSize', 11);
title(['ROC Curve - HAPPY (AUC = ' num2str(AUC_happy, '%.2f') ')'], 'FontSize',
12, 'FontWeight', 'bold');
grid on;

%% --- 6. חישוב מדדי רגישות וסגוליות והדפסתם לחלון הפקודות ---
cp = confusionmat(Y_actual, predicted_labels, 'Order', {'Happy', 'Sad'});
TP = cp(2,2);
TN = cp(1,1);
FP = cp(1,2);
FN = cp(2,1);

sensitivity = (TP / max(1, (TP + FN))) * 100;
specificity = (TN / max(1, (TN + FP))) * 100;

fprintf('\n===== \n');
fprintf('          תוצאות הערכת המסווג (SVM)          \n');
fprintf('===== \n');
fprintf('Sensitivity 2.%(רגישות - עצב):f%%\n', sensitivity);
fprintf('Specificity 2.%(סגוליות - שמחה):f%%\n', specificity);
fprintf('AUC עבור מצב עצב (Sad): %.2f\n', AUC_sad);
fprintf('AUC עבור מצב שמחה (Happy): %.2f\n', AUC_happy);
fprintf('===== \n');

```